

TRAINING REGULATIONS

AUTOMOTIVE DIAGNOSIS (CHASSIS) NC III



AUTOMOTIVE AND LAND TRANSPORTATION SECTOR
TECHNICAL EDUCATION AND SKILLS DEVELOPMENT AUTHORITY
TESDA Complex East Service Road, South Luzon Expressway (SLEX),
Fort Bonifacio, Taguig City

Technical Education and Skills Development Act of 1994
(Republic Act No. 7796)

Section 22, “Establishment and Administration of the National Trade Skills Standards” of the RA 7796 known as the TESDA Act mandates TESDA to establish national occupational skill standards. The Authority shall develop and implement a certification and accreditation program in which private industry group and trade associations are accredited to conduct approved trade tests, and the local government units to promote such trade testing activities in their respective areas in accordance with the guidelines to be set by the Authority.

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The Training Regulations (TR) serve as basis for the:

1. Competency assessment and certification;
2. Registration and delivery of training programs; and
3. Development of curriculum and assessment instruments.

Each TR has four sections:

- | | |
|-----------|---|
| Section 1 | Definition of Qualification - describes the qualification and defines the competencies that comprise the qualification. |
| Section 2 | Competency Standards - was revised to include the Required Knowledge and Required Skills per element. These fields explicitly state the required knowledge and skills for competent performance of a unit of competency in an informed and effective manner. These also emphasize the application of knowledge and skills to situations where understanding is converted into a workplace outcome. |
| Section 3 | Training Arrangements - contain the information and requirements which serve as bases for training providers in designing and delivering competency-based curriculum for the qualification. The revisions to Section 3 entail identifying the Learning Activities leading to achievement of the identified Learning Outcome. |
| Section 4 | Assessment and Certification Arrangements - describe the policies governing assessment and certification procedures for the qualification. |

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TRAINING REGULATIONS FOR

AUTOMOTIVE DIAGNOSIS (CHASSIS) NC III

SECTION 1 DEFINITION OF QUALIFICATION

The **AUTOMOTIVE DIAGNOSIS (CHASSIS) NC III** Qualification consists of competencies that a person must achieve to diagnose, inspect and repair the electronic controlled transmission, continuously variable transmission, vehicle dynamics integrated management, electronic 4-wheel drive control system, electric power steering system.

This Qualification is packaged from the competency map of the Automotive and Land Transportation Industry (Service sector) as shown in Annex A.

The Units of Competency comprising this qualification include the following

CODE NO	BASIC COMPETENCIES
400311319	Lead workplace communication
400311320	Lead small teams
400311321	Apply critical thinking and problem-solving techniques in the workplace
400311322	Work in a diverse environment
400311323	Propose methods of applying learning and innovation in the organization
400311324	Use information systematically
400311325	Evaluate occupational safety and health work practices
400311326	Evaluate environmental work practices
400311327	Facilitate entrepreneurial skills for micro-small-medium enterprises (MSMEs)
CODE NO	COMMON COMPETENCIES
ALT723211	Validate vehicle specification
ALT832212	Move and position vehicle
ALT723213	Utilize automotive tools
ALT723214	Perform mensuration and calculation
ALT723215	Utilize workshop facilities and equipment
ALT723216	Prepare servicing parts and consumables
ALT723217	Prepare vehicle for servicing and releasing
CODE NO	CORE COMPETENCIES
ALT723321	Diagnose and repair of electronically controlled transmission
ALT723322	Diagnose and replace of continuously variable transmission
ALT723365	Diagnose and repair of vehicle dynamics integrated management
ALT723366	Diagnose and repair of electronic 4-wheel drive control system
ALT723367	Diagnose and repair 4-wheel drive assembly
ALT723368	Diagnose and repair electric power steering

A person who has achieved this Qualification is competent to be:

- Automotive Chassis Diagnosis Specialist

SECTION 2 COMPETENCY STANDARDS

This section gives the details of the contents of the basic, common and core units of competency required in **AUTOMOTIVE DIAGNOSIS (CHASSIS) NC III**.

BASIC COMPETENCIES

UNIT OF COMPETENCY : LEAD WORKPLACE COMMUNICATION

UNIT CODE : 400311319

UNIT DESCRIPTOR : This unit covers the knowledge, skills and attitudes required to lead in the effective dissemination and discussion of ideas, information, and issues in the workplace. This includes preparation of written communication materials.

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Communicate information about workplace processes	1.1 Relevant <i>communication method</i> is selected based on workplace procedures 1.2 Multiple operations involving several topics/areas are communicated following enterprise requirements 1.3 Questioning is applied to gain extra information 1.4 Relevant sources of information are identified in accordance with workplace/ client requirements 1.5 Information is selected and organized following enterprise procedures 1.6 Verbal and written reporting is	1.1. Organization requirements for written and electronic communication methods 1.2. Effective verbal communication methods 1.3. Business writing 1.4. Workplace etiquette	1.1 Organizing information 1.2 Conveying intended meaning 1.3 Participating in a variety of workplace discussions 1.4 Complying with organization requirements for the use of written and electronic communication methods 1.5 Effective business writing 1.6 Effective clarifying and probing skills 1.7 Effective questioning techniques (clarifying and probing)

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
	undertaken when required 1.7 Communication and negotiation skills are applied and maintained in all relevant situations		
2. Lead workplace discussions	2.1 Response to workplace issues are sought following enterprise procedures 2.2 Response to workplace issues are provided immediately 2.3 Constructive contributions are made to <i>workplace discussions</i> on such issues as production, quality and safety 2.4 Goals/ objectives and action plans undertaken in the workplace are communicated promptly	2.1 Organization requirements for written and electronic communication methods 2.2 Effective verbal communication methods 2.3 Workplace etiquette	2.1 Organizing information 2.2 Conveying intended meaning 2.3 Participating in variety of workplace discussions 2.4 Complying with organization requirements for the use of written and electronic communication methods 2.5 Effective clarifying and probing skills
3. Identify and communicate issues arising in the workplace	3.1 Issues and problems are identified as they arise 3.2 Information regarding problems and issues are organized coherently to ensure clear and effective communication 3.3 Dialogue is initiated with appropriate personnel 3.4 Communication problems and issues are raised as they arise	3.1 Organization requirements for written and electronic communication methods 3.2 Effective verbal communication methods 3.3 Workplace etiquette 3.4 Communication problems and issues 3.5 Barriers in communication	3.1 Organizing information 3.2 Conveying intended meaning 3.3 Participating in a variety of workplace discussions 3.4 Complying with organization requirements for the use of written and electronic communication methods 3.5 Effective clarifying and probing skills 3.6 Identifying issues

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
	3.5 Identify barriers in communication to be addressed appropriately		3.7 Negotiation and communication skills

RANGE OF VARIABLES

VARIABLE	RANGE
1. Methods of communication	May include: 1.1. Non-verbal gestures 1.2. Verbal 1.3. Face-to-face 1.4. Two-way radio 1.5. Speaking to groups 1.6. Using telephone 1.7. Written 1.8. Internet
2. Workplace discussions	May include: 2.1. Coordination meetings 2.2. Toolbox discussion 2.3. Peer-to-peer discussion

EVIDENCE GUIDE

1. Critical aspects of Competency	Assessment requires evidence that the candidate: 1.1 Dealt with a range of communication/information at one time 1.2 Demonstrated leadership skills in workplace communication 1.3 Made constructive contributions in workplace issues 1.4 Sought workplace issues effectively 1.5 Responded to workplace issues promptly 1.6 Presented information clearly and effectively written form 1.7 Used appropriate sources of information 1.8 Asked appropriate questions 1.9 Provided accurate information
2. Resource Implications	The following resources should be provided: 2.1 Variety of Information 2.2 Communication tools 2.3 Simulated workplace
3. Methods of Assessment	Competency in this unit may be assessed through: Case problem 3.1. Third-party report 3.2. Portfolio 3.3. Interview 3.4. Demonstration/Role-playing
4. Context for Assessment	4.1. Competency may be assessed in the workplace or in a simulated workplace environment

UNIT OF COMPETENCY : LEAD SMALL TEAMS

UNIT CODE : 400311320

UNIT DESCRIPTOR : This unit covers the knowledge, skills and attitudes to lead small teams including setting, maintaining and monitoring team and individual performance standards.

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Provide team leadership	1.1 <i>Work requirements</i> are identified and presented to team members based on company policies and procedures 1.2 Reasons for instructions and requirements are communicated to team members based on company policies and procedures 1.3 <i>Team members' and leaders' concerns</i> are recognized, discussed and dealt with based on company practices	1.1 Facilitation of Team work 1.2 Company policies and procedures relating to work performance 1.3 Performance standards and expectations 1.4 Monitoring individual's and team's performance vis a vis client's and group's expectations	1.1 Communication skills required for leading teams 1.2 Group facilitation skills 1.3 Negotiating skills 1.4 Setting performance expectation
2. Assign responsibilities	2.1. Responsibilities are allocated having regard to the skills, knowledge and aptitude required to undertake the assigned task based on company policies. 2.2. Duties are allocated having regard to individual preference, domestic and personal considerations, whenever possible	2.1 Work plan and procedures 2.2 Work requirements and targets 2.2 Individual and group expectations and assignments 2.3 Ways to improve group leadership and membership	2.1 Communication skills 2.2 Management skills 2.3 Negotiating skills 2.4 Evaluation skills 2.5 Identifying team member's strengths and rooms for improvement

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
3. Set performance expectations for team members	3.1 Performance expectations are established based on client needs 3.2 Performance expectations are based on individual team members knowledge, skills and aptitude 3.3 Performance expectations are discussed and disseminated to individual team members	3.1 One's roles and responsibilities in the team 3.2 Feedback giving and receiving 3.3 Performance expectation	3.1 Communication skills 3.2 Accurate empathy 3.3 Congruence 3.4 Unconditional positive regard 3.5 Handling of Feedback
4. Supervise team performance	4.1 Performance is monitored based on defined performance criteria and/or assignment instruction 4.2 Team members are provided with <i>feedback</i> , positive support and advice on strategies to overcome any deficiencies based on company practices 4.3 <i>Performance issues</i> which cannot be rectified or addressed within the team are referred to appropriate personnel according to employer policy 4.4 Team members are kept informed of any changes in the priority allocated to assignments or tasks which might impact on	4.1 Performance Coaching 4.2 Performance management 4.3 Performance Issues	4.1 Communication skills required for leading teams 4.2 Coaching skill

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
	client/customer needs and satisfaction 4.5 Team operations are monitored to ensure that employer/client needs and requirements are met 4.6 Follow-up communication is provided on all issues affecting the team 4.7 All relevant documentation is completed in accordance with company procedures		
5. Set performance expectations for team members	5.1 Performance expectations are established based on client needs 5.2 Performance expectations are based on individual team members knowledge, skills and aptitude 5.3 Performance expectations are discussed and disseminated to individual team members	5.1 One's roles and responsibilities in the team 5.2 Feedback giving and receiving 5.3 Performance expectation	5.1 Communication skills 5.2 Accurate empathy 5.3 Congruence 5.4 Unconditional positive regard 5.5 Handling of Feedback
6. Supervise team performance	6.1 Performance is monitored based on defined performance criteria and/or assignment instruction 6.2 Team members are provided with <i>feedback</i> , positive support and advice on strategies to	6.1 Performance Coaching 6.2 Performance management 6.3 Performance Issues	6.1 Communication skills required for leading teams 6.2 Coaching skill

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
	overcome any deficiencies based on company practices 6.3 <i>Performance issues</i> which cannot be rectified or addressed within the team are referred to appropriate personnel according to employer policy 6.4 Team members are kept informed of any changes in the priority allocated to assignments or tasks which might impact on client/customer needs and satisfaction 6.5 Team operations are monitored to ensure that employer/client needs and requirements are met 6.6 Follow-up communication is provided on all issues affecting the team 6.7 All relevant documentation is completed in accordance with company procedures		

RANGE OF VARIABLES

VARIABLE	RANGE
1. Work requirements	Work requirements may include: 1.1. Client Profile 1.2. Assignment instructions
2. Team member's concerns	Team member's concerns may include: 2.1. Roster/shift details
3. Monitor performance	Monitor performance may include: 3.1. Formal process 3.2. Informal process
4. Feedback	Feedback may include: 4.1. Formal process 4.2. Informal process
5. Performance issues	Performance issues may include: 5.1. Work output 5.2. Work quality 5.3. Team participation 5.4. Compliance with workplace protocols 5.5. Safety 5.6. Customer service

EVIDENCE GUIDE

1. Critical aspects of Competency	Assessment requires evidence that the candidate: 1.1. Maintained or improved individuals and/or team performance given a variety of possible scenario 1.2. Assessed and monitored team and individual performance against set criteria 1.3. Represented concerns of a team and individual to next level of management or appropriate specialist and to negotiate on their behalf 1.4. Allocated duties and responsibilities, having regard to individual's knowledge, skills and aptitude and the needs of the tasks to be performed 1.5. Set and communicated performance expectations for a range of tasks and duties within the team and provided feedback to team members
2. Resource Implications	The following resources should be provided: 2.1. Access to relevant workplace or appropriately simulated environment where assessment can take place 2.2. Materials relevant to the proposed activity or task
3. Methods of Assessment	Competency in this unit may be assessed through: 3.1. Written Examination 3.2. Oral Questioning 3.3. Portfolio
4. Context for Assessment	4.1 Competency may be assessed in actual workplace or at the designated TESDA Accredited Assessment Center.

UNIT OF COMPETENCY : APPLY CRITICAL THINKING AND PROBLEM-SOLVING TECHNIQUES IN THE WORKPLACE

UNIT CODE : 400311321

UNIT DESCRIPTOR : This unit covers the knowledge, skills and attitudes required to solve problems in the workplace including the application of problem solving techniques and to determine and resolve the root cause/s of specific problems in the workplace.

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Examine specific workplace challenges.	1.1 Variances are examined from normal operating <i>parameters</i> ; and product quality. 1.2 Extent, cause and nature of the specific problem are defined through observation, investigation and <i>analytical techniques</i> . 1.3 <i>Problems</i> are clearly stated and specified.	1.1 Competence includes a thorough knowledge and understanding of the process, normal operating parameters, and product quality to recognize non-standard situations. 1.2 Competence to include the ability to apply and explain, enough for the identification of fundamental causes of specific workplace challenges. 1.3 Relevant equipment and operational processes. 1.4 Enterprise goals, targets and measures. 1.5 Enterprise quality OHS and environmental requirement. 1.6 Enterprise information	1.1 Using range of analytical techniques (e.g., planning, attention, simultaneous and successive processing of information) in examining specific challenges in the workplace. 1.2 Identifying extent and causes of specific challenges in the workplace.

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
		systems and data collation 1.7 Industry codes and standards.	
2. Analyze the causes of specific workplace challenges.	2.1 Possible causes of specific problems are identified based on experience and the use of problem solving tools / analytical techniques. 2.2 Possible cause statements are developed based on findings. 2.3 Fundamental causes are identified per results of investigation conducted.	2.1 Competence includes a thorough knowledge and understanding of the process, normal operating parameters, and product quality to recognize non-standard situations. 2.2 Competence to include the ability to apply and explain, sufficient for the identification of fundamental cause, determining the corrective action and provision of recommendations. 2.3 Relevant equipment and operational processes. 2.4 Enterprise goals, targets and measures. 2.5 Enterprise quality OSH and environmental requirement. 2.6 Enterprise information systems and data collation. 2.7 Industry codes and standards.	2.1 Using range of analytical techniques (e.g., planning, attention, simultaneous and successive processing of information) in examining specific challenges in the workplace. 2.2 Identifying extent and causes of specific challenges in the workplace. 2.3 Providing clear-cut findings on the nature of each identified workplace challenges.
3. Formulate resolutions to	3.1 All possible options are considered for	3.1 Competence to include the ability to apply and	3.1 Using range of analytical techniques (e.g.,

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
specific workplace challenges.	resolution of the problem. 3.2 Strengths and weaknesses of possible options are considered. 3.3 Corrective actions are determined to resolve the problem and possible future causes. 3.4 Action plans are developed identifying measurable objectives, resource needs and timelines in accordance with safety and operating procedures	explain, sufficient for the identification of fundamental cause, determining the corrective action and provision of recommendations 3.2 Relevant equipment and operational processes 3.3 Enterprise goals, targets and measures 3.4 Enterprise quality OSH and environmental requirement 3.5 Principles of decision making strategies and techniques 3.6 Enterprise information systems and data collation 3.7 Industry codes and standards	planning, attention, simultaneous and successive processing of information) in examining specific challenges in the workplace. 3.2 Identifying extent and causes of specific challenges in the workplace. 3.3 Providing clear-cut findings on the nature of each identified workplace challenges. 2.4 Devising, communicating, implementing and evaluating strategies and techniques in addressing specific workplace challenges.
4. Implement action plans and communicate results.	4.1 Action plans are implemented and evaluated. 4.2 Results of plan implementation and recommendations are prepared. 4.2 Recommendations are presented to appropriate personnel. 4.3 Recommendations are followed-up, if required.	4.1 Competence to include the ability to apply and explain, sufficient for the identification of fundamental cause, determining the corrective action and provision of recommendations 4.2. Relevant equipment and operational processes	4.1 Using range of analytical techniques (e.g., planning, attention, simultaneous and successive processing of information) in examining specific challenges in the workplace. 4.2 Identifying extent and causes of specific

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
		4.3 Enterprise goals, targets and measures 4.4 Enterprise quality, OSH and environmental requirement 4.5 Principles of decision making strategies and techniques 4.6 Enterprise information systems and data collation 4.7 Industry codes and standards	challenges in the workplace. 4.3 Providing clear-cut findings on the nature of each identified workplace challenges. 4.4 Devising, communicating, implementing and evaluating strategies and techniques in addressing specific workplace challenges.

RANGE OF VARIABLES

VARIABLE	RANGE
1. Parameters	Parameters may include: 1.1 Processes 1.2 Procedures 1.3 Systems
2. Analytical techniques	Analytical techniques may include: 2.1. Brainstorming 2.2. Intuitions/Logic 2.3. Cause and effect diagrams 2.4. Pareto analysis 2.5. SWOT analysis 2.6. Gant chart, Pert CPM and graphs 2.7. Scattergrams
3. Problem	Problem may include: 3.1. Routine, non – routine and complex workplace and quality problems 3.2. Equipment selection, availability and failure 3.3. Teamwork and work allocation problem 3.4. Safety and emergency situations and incidents 3.5. Risk assessment and management
4. Action plans	Action plans may include: 4.1. Priority requirements 4.2. Measurable objectives 4.3. Resource requirements 4.4. Timelines 4.5. Co-ordination and feedback requirements 4.6. Safety requirements 4.7. Risk assessment 4.8. Environmental requirements

EVIDENCE GUIDE

1. Critical aspects of Competency	Assessment requires evidence that the candidate: 1.1. Examined specific workplace challenges. 1.2. Analyzed the causes of specific workplace challenges. 1.3. Formulated resolutions to specific workplace challenges. 1.4. Implemented action plans and communicated results on specific workplace challenges.
2. Resource Implications	2.1. Assessment will require access to an operating plant over an extended period of time, or a suitable method of gathering evidence of operating ability over a range of situations. A bank of scenarios / case studies / what ifs will be required as well as bank of questions which will be used to probe the reason behind the observable action.
3. Methods of Assessment	Competency in this unit may be assessed through: 3.1. Observation 3.2. Case Formulation 3.3. Life Narrative Inquiry 3.4. Standardized test The unit will be assessed in a holistic manner as is practical and may be integrated with the assessment of other relevant units of competency. Assessment will occur over a range of situations, which will include disruptions to normal, smooth operation. Simulation may be required to allow for timely assessment of parts of this unit of competency. Simulation should be based on the actual workplace and will include walk through of the relevant competency components. These assessment activities should include a range of problems, including new, unusual and improbable situations that may have happened.
4. Context for Assessment	In all workplace, it may be appropriate to assess this unit concurrently with relevant teamwork or operation units.

UNIT OF COMPETENCY : WORK IN A DIVERSE ENVIRONMENT

UNIT CODE : 400311322

UNIT DESCRIPTOR : This unit covers the outcomes required to work effectively in a workplace characterized by diversity in terms of religions, beliefs, races, ethnicities and other differences.

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Develop an individual's cultural awareness and sensitivity	1.1 Individual differences with clients, customers and fellow workers are recognized and respected in accordance with enterprise policies and core values. 1.2 Differences are responded to in a sensitive and considerate manner 1.3 Diversity is accommodated using appropriate verbal and non-verbal communication.	1.1 Understanding cultural diversity in the workplace 1.2 Norms of behavior for interacting and dialogue with specific groups (e. g., Muslims and other non-Christians, non-Catholics, tribes/ethnic groups, foreigners) 1.3 Different methods of verbal and non-verbal communication in a multicultural setting	1.1 Applying cross-cultural communication skills (i.e. different business customs, beliefs, communication strategies) 1.2 Showing affective skills – establishing rapport and empathy, understanding, etc. 1.3 Demonstrating openness and flexibility in communication 1.4 Recognizing diverse groups in the workplace and community as defined by divergent culture, religion, traditions and practices
2. Work effectively in an environment that acknowledges and values cultural diversity	2.1 Knowledge, skills and experiences of others are recognized and documented in relation to team objectives.	2.1 Value of diversity in the economy and society in terms of Workforce development 2.2 Importance of inclusiveness in a	2.1 Demonstrating cross-cultural communication skills and active listening 2.2 Recognizing diverse groups in the workplace

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
	2.2 Fellow workers are encouraged to utilize and share their specific qualities, skills or backgrounds with other team members and clients to enhance work outcomes. 2.3 Relations with customers and clients are maintained to show that diversity is valued by the business.	diverse environment 2.3 Shared vision and understanding of and commitment to team, departmental, and organizational goals and objectives 2.4 Strategies for customer service excellence	and community as defined by divergent culture, religion, traditions and practices 2.3 Demonstrating collaboration skills 2.4 Exhibiting customer service excellence
3. Identify common issues in a multicultural and diverse environment	3.1 <i>Diversity-related conflicts</i> within the workplace are effectively addressed and resolved. 3.2 Discriminatory behaviors towards customers/stakeholders are minimized and addressed accordingly. 3.3 Change management policies are in place within the organization.	3.1 Value, and leverage of cultural diversity 3.2 Inclusivity and conflict resolution 3.3 Workplace harassment 3.4 Change management and ways to overcome resistance to change 3.5 Advanced strategies for customer service excellence	3.1 Addressing diversity-related conflicts in the workplace 3.2 Eliminating discriminatory behavior towards customers and co-workers 3.3 Utilizing change management policies in the workplace

RANGE OF VARIABLES

VARIABLE	RANGE
1. Diversity	This refers to diversity in both the workplace and the community and may include divergence in : 1.1 Religion 1.2 Ethnicity, race or nationality 1.3 Culture 1.4 Gender, age or personality 1.5 Educational background
2. Diversity-related conflicts	May include conflicts that result from: 2.1 Discriminatory behaviors 2.2 Differences of cultural practices 2.3 Differences of belief and value systems 2.4 Gender-based violence 2.5 Workplace bullying 2.6 Corporate jealousy 2.7 Language barriers 2.8 Individuals being differently-abled persons 2.9 Ageism (negative attitude and behavior towards old people)

EVIDENCE GUIDE

1. Critical aspects of Competency	Assessment requires evidence that the candidate: 1.1 Adjusted language and behavior as required by interactions with diversity 1.2 Identified and respected individual differences in colleagues, clients and customers 1.3 Applied relevant regulations, standards and codes of practice
2. Resource Implications	The following resources should be provided: 2.1 Access to workplace and resources 2.2 Manuals and policies on Workplace Diversity
3. Methods of Assessment	Competency in this unit may be assessed through: 3.1 Demonstration or simulation with oral questioning 3.2 Group discussions and interactive activities 3.3 Case studies/problems involving workplace diversity issues 3.4 Third-party report 3.5 Written examination 3.6 Role Plays
4. Context for Assessment	Competency assessment may occur in workplace or any appropriately simulated environment

UNIT OF COMPETENCY : PROPOSE METHODS OF APPLYING LEARNING AND INNOVATION IN THE ORGANIZATION

UNIT CODE : 400311323

UNIT DESCRIPTOR : This unit covers the knowledge, skills and attitudes required to assess general obstacles in the application of learning and innovation in the organization and to propose practical methods of such in addressing organizational challenges.

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Assess work procedures, processes and systems in terms of innovative practices.	1.1. <i>Reasons</i> for innovation are incorporated to work procedures. 1.2. <i>Models of innovation</i> are researched. 1.3. <i>Gaps or barriers</i> to innovation in one's work area are analyzed. 1.4. Staff who can support and foster innovation in the work procedure are identified.	1.1 Seven habits of highly effective people. 1.2 Character strengths that foster innovation and learning (Christopher Peterson and Martin Seligman, 2004) 1.3 Five minds of the future concepts (Gardner, 2007). 1.4 Adaptation concepts in neuroscience (Merzenich, 2013). 1.5 Transtheoretical model of behavior change (Prochaska, DiClemente, & Norcross, 1992).	1.1 Demonstrating collaboration and networking skills. 1.2 Applying basic research and evaluation skills 1.3 Generating insights on how to improve organizational procedures, processes and systems through innovation.

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
2. Generate practical action plans for improving work procedures, processes.	2.1 Ideas for innovative work procedure to foster innovation using individual and group techniques are conceptualized 2.2 Range of ideas with other team members and colleagues are evaluated and discussed 2.3 Work procedures and processes subject to change are selected based on workplace requirements (feasible and innovative). 2.4 Practical action plans are proposed to facilitate simple changes in the work procedures, processes and systems. 2.5 Critical inquiry is applied and used to facilitate discourse on adjustments in the simple work procedures, processes and systems.	2.1 Seven habits of highly effective people. 2.2 Character strengths that foster innovation and learning (Christopher Peterson and Martin Seligman, 2004) 2.3 Five minds of the future concepts (Gardner, 2007). 2.4 Adaptation concepts in neuroscience (Merzenich, 2013). 2.5 Transtheoretical model of behavior change (Prochaska, DiClemente, & Norcross, 1992).	2.1 Assessing readiness for change on simple work procedures, processes and systems. 2.2 Generating insights on how to improve organizational procedures, processes and systems through innovation. 2.3 Facilitating action plans on how to apply innovative procedures in the organization.

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
3. Evaluate the effectiveness of the proposed action plans.	3.1 Work structure is analyzed to identify the impact of the new work procedures 3.2 Co-workers/key personnel is consulted to know who will be involved with or affected by the work procedure 3.3 Work instruction operational plan of the new work procedure is developed and evaluated. 3.4 Feedback and suggestion are recorded. 3.5 Operational plan is updated. 3.6 Results and impact on the developed work instructions are reviewed 3.7 Results of the new work procedure are evaluated 3.8 Adjustments are recommended based on results gathered	3.1 Five minds of the future concepts (Gardner, 2007). 3.2 Adaptation concepts in neuroscience (Merzenich, 2013). 3.3 Transtheoretical model of behavior change (Prochaska, DiClemente, & Norcross, 1992).	3.1 Generating insights on how to improve organizational procedures, processes and systems through innovation. 3.2 Facilitating action plans on how to apply innovative procedures in the organization. 3.3 Communicating results of the evaluation of the proposed and implemented changes in the workplace procedures and systems. 3.4 Developing action plans for continuous improvement on the basic systems, processes and procedures in the organization.

RANGE OF VARIABLES

VARIABLE	RANGE
1. Reasons	May include: 1.1 Strengths and weaknesses of the current systems, processes and procedures. 1.2 Opportunities and threats of the current systems, processes and procedures.
2. Models of innovation	May include: 2.1 Seven habits of highly effective people. 2.2 Five minds of the future concepts (Gardner, 2007). 2.3 Neuroplasticity and adaptation strategies.
3. Gaps or barriers	May include: 3.1 Machine 3.2 Manpower 3.3 Methods 3.4 Money
4. Critical Inquiry	May include: 4.1 Preparation. 4.2 Discussion. 4.3 Clarification of goals. 4.4 Negotiate towards a Win-Win outcome. 4.5 Agreement. 4.6 Implementation of a course of action. 4.7 Effective verbal communication. See our pages: Verbal Communication and Effective Speaking. 4.8 Listening. 4.9 Reducing misunderstandings is a key part of effective negotiation. 4.10 Rapport Building. 4.11 Problem Solving. 4.12 Decision Making. 4.13 Assertiveness. 4.14 Dealing with Difficult Situations.

EVIDENCE GUIDE

1. Critical aspects of Competency	Assessment requires evidence that the candidate: 1.1 Established the reasons why innovative systems are required 1.2 Established the goals of a new innovative system 1.3 Analyzed current organizational systems to identify gaps and barriers to innovation. 1.4 Assessed work procedures, processes and systems in terms of innovative practices. 1.5 Generate practical action plans for improving work procedures, and processes. 1.6 Reviewed the trial innovative work system and adjusted reflect evaluation feedback, knowledge management systems and future planning. 1.7 Evaluated the effectiveness of the proposed action plans.
2. Resource Implications	The following resources should be provided: 2.1 Pens, papers and writing implements. 2.2 Cartolina. 2.3 Manila papers.
3. Methods of Assessment	Competency in this unit may be assessed through: 3.1 Psychological and behavioral Interviews. 3.2 Performance Evaluation. 3.3 Life Narrative Inquiry. 3.4 Review of portfolios of evidence and third-party workplace reports of on-the-job performance. 3.5 Sensitivity analysis. 3.6 Organizational analysis. 3.7 Standardized assessment of character strengths and virtues applied.
4. Context for Assessment	4.1 Competency may be assessed individually in the actual workplace or simulation environment in TESDA accredited institutions.

UNIT OF COMPETENCY : USE INFORMATION SYSTEMATICALLY

UNIT CODE : 400311324

UNIT DESCRIPTOR : This unit covers the knowledge, skills and attitudes required to use technical information systems, apply information technology (IT) systems and edit, format and check information.

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Use technical information	1.1. Information are collated and organized into a suitable form for reference and use 1.2. Stored information are classified so that it can be quickly identified and retrieved when needed 1.3. Guidance are advised and offered to people who need to find and use information	1.1. Application in collating information 1.2. Procedures for inputting, maintaining and archiving information 1.3. Guidance to people who need to find and use information 1.4. Organize information 1.5. classify stored information for identification and retrieval 1.6. Operate the technical information system by using agreed procedures	1.1. Collating information 1.2. Operating appropriate and valid procedures for inputting, maintaining and archiving information 1.3. Advising and offering guidance to people who need to find and use information 1.4. Organizing information into a suitable form for reference and use 1.5. Classifying stored information for identification and retrieval 1.6. Operating the technical information system by using agreed procedures

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
2. Apply information technology (IT)	2.1. Technical information system is operated using agreed procedures 2.2. Appropriate and valid procedures are operated for inputting, maintaining and archiving information 2.3. Software required are utilized to execute the project activities 2.4. Information and data obtained are handled, edited, formatted and checked from a range of internal and external sources 2.5. Information are extracted, entered, and processed to produce the outputs required by customers 2.6. Own skills and understanding are shared to help others 2.7. Specified security measures are implemented to protect the confidentiality and integrity of project data held in IT systems	2.1. Attributes and limitations of available software tools 2.2. Procedures and work instructions for the use of IT 2.3. Operational requirements for IT systems 2.4. Sources and flow paths of data 2.5. Security systems and measures that can be used 2.6. Extract data and format reports 2.7. Methods of entering and processing information 2.8. WWW enabled applications	2.1. Identifying attributes and limitations of available software tools 2.2. Using procedures and work instructions for the use of IT 2.3. Describing operational requirements for IT systems 2.4. Identifying sources and flow paths of data 2.5. Determining security systems and measures that can be used 2.6. Extracting data and format reports 2.7. Describing methods of entering and processing information 2.8. Using WWW applications

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
3. Edit, format and check information	3.1 Basic editing techniques are used 3.2 Accuracy of documents are checked 3.3 Editing and formatting tools and techniques are used for more complex documents 3.4 Proofreading techniques is used to check that documents look professional	3.1 Basic file-handling techniques 3.2 Techniques in checking documents 3.3 Techniques in editing and formatting 3.4 Proof reading techniques	3.1 Using basic file-handling techniques is used for the software 3.2 Using different techniques in checking documents 3.3 Applying editing and formatting techniques 3.4 Applying proofreading techniques

RANGE OF VARIABLES

VARIABLE	RANGE
1. Information	May include: 1.1. Property 1.2. Organizational 1.3. Technical reference
2. Technical information	May include: 2.1. paper based 2.2. electronic
3. Software	May include: 3.1. spreadsheets 3.2. databases 3.3. word processing 3.4. presentation
4. Sources	May include: 4.1. other IT systems 4.2. manually created 4.3. within own organization 4.4. outside own organization 4.5. geographically remote
5. Customers	May include: 5.1. colleagues 5.2. company and project management 5.3. clients
6. Security measures	May include: 6.1. access rights to input; 6.2. passwords; 6.3. access rights to outputs; 6.4. data consistency and back-up; 6.5. recovery plans

EVIDENCE GUIDE

1. Critical aspects of Competency	Assessment requires evidence that the candidate: 1.1. Used technical information systems and information technology 1.2. Applied information technology (IT) systems 1.3. Edited, formatted and checked information
2. Resource Implications	The following resources <u>MUST</u> be provided: 2.1. Computers 2.2. Software and IT system
3. Methods of Assessment	Competency in this unit <u>MUST</u> be assessed through: 3.1. Direct Observation 3.2. Oral interview and written test
4. Context for Assessment	4.1. Competency may be assessed individually in the actual workplace or through accredited institution

UNIT OF COMPETENCY : EVALUATE OCCUPATIONAL SAFETY AND HEALTH WORK PRACTICES

UNIT CODE : 400311325

UNIT DESCRIPTOR : This unit covers the knowledge, skills and attitudes required to interpret Occupational Safety and Health practices, set OSH work targets, and evaluate effectiveness of Occupational Safety and Health work instructions.

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Interpret Occupational Safety and Health practices	1.1 <i>OSH work practices issues</i> are identified relevant to work requirements 1.2 OSH work standards and procedures are determined based on applicability to nature of work 1.3 Gaps in work practices are identified related to relevant OSH work standards	1.1. OSH work practices issues 1.2. OSH work standards 1.3. General OSH principles and legislations 1.4. Company/ workplace policies/ guidelines 1.5. Standards and safety requirements of work process and procedures	1.1. Communication skills 1.2. Interpersonal skills 1.3. Critical thinking skills 1.4. Observation skills
2. Set OSH work targets	2.1 Relevant work information are gathered necessary to determine OSH work targets 2.2 <i>OSH Indicators</i> based on gathered information are agreed upon to measure effectiveness of workplace OSH policies and procedures 2.3 Agreed OSH indicators are endorsed for approval from appropriate personnel 2.4 <i>OSH work instructions</i> are	2.1. OSH work targets 2.2. OSH Indicators 2.3. OSH work instructions 2.4. Safety and health requirements of tasks 2.5. Workplace guidelines on providing feedback on OSH and security concerns 2.6. OSH regulations Hazard control procedures 2.7. OSH trainings relevant to work	2.1. Communication skills 2.2. Collaborating skills 2.3. Critical thinking skills 2.4. Observation skills

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
	received in accordance with workplace policies and procedures*		
3. Evaluate effectiveness of Occupational Safety and Health work instructions	3.1 OSH Practices are observed based on workplace standards 3.2 Observed OSH practices are measured against approved OSH metrics 3.3 Findings regarding effectiveness are assessed and gaps identified are implemented based on OSH work standards	3.1. OSH Practices 3.2. OSH metrics 3.3. OSH Evaluation Techniques 3.4. OSH work standards	3.1. Critical thinking skills 3.2. Evaluating skills

RANGE OF VARIABLES

VARIABLE	RANGE
1. OSH Work Practices Issues	May include: 1.1 Workers' experience/observance on presence of work hazards 1.2 Unsafe/unhealthy administrative arrangements (prolonged work hours, no break-time, constant overtime, scheduling of tasks) 1.3 Reasons for compliance/non-compliance to use of PPEs or other OSH procedures/policies/ guidelines
2. OSH Indicators	May include: 2.1 Increased of incidents of accidents, injuries 2.2 Increased occurrence of sickness or health complaints/symptoms 2.3 Common complaints of workers' related to OSH 2.4 High absenteeism for work-related reasons
3. OSH Work Instructions	May include: 3.1 Preventive and control measures, and targets 3.2 Eliminate the hazard (i.e., get rid of the dangerous machine 3.3 Isolate the hazard (i.e. keep the machine in a closed room and operate it remotely; barricade an unsafe area off) 3.4 Substitute the hazard with a safer alternative (i.e., replace the machine with a safer one) 3.5 Use administrative controls to reduce the risk (i.e. give trainings on how to use equipment safely; OSH-related topics, issue warning signages, rotation/shifting work schedule) 3.6 Use engineering controls to reduce the risk (i.e. use safety guards to machine) 3.7 Use personal protective equipment 3.8 Safety, Health and Work Environment Evaluation 3.9 Periodic and/or special medical examinations of workers
4. OSH metrics	May include: 4.1 Statistics on incidence of accident and injuries 4.2 Morbidity (Type and Number of Sickness) 4.3 Mortality (Cause and Number of Deaths) 4.4 Accident Rate

EVIDENCE GUIDE

1. Critical aspects of Competency	Assessment requires evidence that the candidate: 1.1. Identify OSH work practices issues relevant to work requirements 1.2. Identify gaps in work practices related to relevant OSH work standards 1.3. Agree upon OSH Indicators based on gathered information to measure effectiveness of workplace OSH policies and procedures 1.4. Receive OSH work instructions in accordance with workplace policies and procedures 1.5. Compare Observed OSH practices with against approved OSH work instructions 1.6. Assess findings regarding effectiveness based on OSH work standards
2. Resource Implications	The following resources should be provided: 2.1 Facilities, materials, tools and equipment necessary for the activity
3. Methods of Assessment	Competency in this unit may be assessed through: 3.1 Observation/Demonstration with oral questioning 3.2 Third party report 3.3 Written exam
4. Context for Assessment	4.1 Competency may be assessed in the work place or in a simulated work place setting

UNIT OF COMPETENCY : EVALUATE ENVIRONMENTAL WORK PRACTICES

UNIT CODE : 400311326

UNIT DESCRIPTOR : This unit covers the knowledge, skills and attitude to interpret environmental Issues, establish targets to evaluate environmental practices and evaluate effectiveness of environmental practices.

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Interpret environmental practices, policies and procedures	1.1 <i>Environmental work practices</i> issues are identified relevant to work requirements 1.2 Environmental Standards and Procedures nature of work are determined based on Applicability to nature of work 1.3 Gaps in work practices related to Environmental Standards and Procedures are identified	1.1 Environmental Issues 1.2 Environmental Work Procedures 1.3 Environmental Laws 1.4 Environmental Hazardous and Non-Hazardous Materials 1.5 Environmental required license, registration or certification	1.1. Analyzing Environmental Issues and Concerns 1.2. Critical thinking 1.3. Problem Solving 1.4. Observation Skills
2. Establish targets to evaluate environmental practices	2.1. Relevant information are gathered necessary to determine environmental work targets 2.2. <i>Environmental Indicators</i> based on gathered information are set to measure environmental work targets 2.3. Indicators are verified with appropriate personnel	2.1. Environmental Indicators 2.2. Relevant Environment Personnel or expert 2.3. Relevant Environmental Trainings and Seminars	2.1. Investigative Skills 2.2. Critical thinking 2.3. Problem Solving 2.4. Observation Skills

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
3. Evaluate effectiveness of environmental practices	3.1. Work environmental practices are recorded based on workplace standards 3.2. Recorded work environmental practices are compared against planned indicators 3.3. Findings regarding effectiveness are assessed and gaps identified are implemented based on environment work standards and procedures 3.4. Results of environmental assessment are conveyed to appropriate personnel	1.1. Environmental Practices 1.2. Environmental Standards and Procedures	3.1 Documentation and Record Keeping Skills 3.2 Critical thinking 3.3 Problem Solving 3.4 Observation Skills

RANGE OF VARIABLES

VARIABLE	RANGE
1. Environmental Practices Issues	May include: 1.1 Water Quality 1.2 National and Local Government Issues 1.3 Safety 1.4 Endangered Species 1.5 Noise 1.6 Air Quality 1.7 Historic 1.8 Waste 1.9 Cultural
2. Environmental Indicators	May include: 2.1 Noise level 2.2 Lighting (Lumens) 2.3 Air Quality - Toxicity 2.4 Thermal Comfort 2.5 Vibration 2.6 Radiation 2.7 Quantity of the Resources 2.8 Volume

EVIDENCE GUIDE

1. Critical aspects of Competency	Assessment requires evidence that the candidate: 1.1. Identified environmental issues relevant to work requirements 1.2. Identified gaps in work practices related to Environmental Standards and Procedures 1.3. Gathered relevant information necessary to determine environmental work targets 1.4. Set environmental indicators based on gathered information to measure environmental work targets 1.5. Recorded work environmental practices are recorded based on workplace standards 1.6. Conveyed results of environmental assessment to appropriate personnel
2. Resource Implications	The following resources should be provided: 2.1 Workplace/Assessment location 2.2 Legislation, policies, procedures, protocols and local ordinances relating to environmental protection 2.3 Case studies/scenarios relating to environmental protection
3. Methods of Assessment	Competency in this unit may be assessed through: 3.1 Written/ Oral Examination 3.2 Interview/Third Party Reports 3.3 Portfolio (citations/awards from GOs and NGOs, certificate of training – local and abroad) 3.4 Simulations and role-plays
4. Context for Assessment	4.1 Competency may be assessed in actual workplace or at the designated TESDA center.

UNIT OF COMPETENCY : FACILITATE ENTREPRENEURIAL SKILLS FOR MICRO-SMALL-MEDIUM ENTERPRISES (MSMEs)

UNIT CODE : 400311327

UNIT DESCRIPTOR : This unit covers the outcomes required to build, operate and grow a micro/small-scale enterprise.

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Develop and maintain micro-small-medium enterprise (MSMEs) skills in the organization	1.1 Appropriate business strategies are determined and set for the enterprise based on current and emerging business environment. 1.2 Business operations are monitored and controlled following established procedures. 1.3 Quality assurance measures are implemented consistently. 1.4 Good relations are maintained with staff/ workers. 1.5 Policies and procedures on occupational safety and health and environmental concerns are constantly observed.	1.1 Business models and strategies 1.2 Types and categories of businesses 1.3 Business operation 1.4 Basic Bookkeeping 1.5 Business internal controls 1.6 Basic quality control and assurance concepts 1.7 Government and regulatory processes	1.1 Basic bookkeeping/ accounting skills 1.2 Communication skills 1.3 Building relations with customer and employees 1.4 Building competitive advantage of the enterprise
2. Establish and Maintain client-base/market	2.1 Good customer relations are maintained 2.2 New customers and markets are identified, explored and reached out to. 2.3 Promotions/ Incentives are	2.1 Public relations concepts 2.2 Basic product promotion strategies 2.3 Basic market and feasibility studies 2.4 Basic business ethics	2.1 Building customer relations 2.2 Individual marketing skills 2.3 Using basic advertising (posters/ tarpaulins, flyers,

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
	offered to loyal customers 2.4 Additional products and services are evaluated and tried where feasible. 2.5 <i>Promotional/advertising initiatives</i> are carried out where necessary and feasible.		social media, etc.)
3. Apply budgeting and financial management skills	3.1 Enterprise is built up and sustained through judicious control of cash flows. 3.2 Profitability of enterprise is ensured through appropriate <i>internal controls</i>. 3.3 Unnecessary or lower-priority expenses and purchases are avoided.	3.1 Cash flow management 3.1 Basic financial management 3.2 Basic financial accounting 3.3 Business internal controls	3.1 Setting business priorities and strategies 3.2 Interpreting basic financial statements 3.3 Preparing business plans

RANGE OF VARIABLES

VARIABLE	RANGE
1. Business strategies	May include: 1.1. Developing/Maintaining niche market 1.2. Use of organic/healthy ingredients 1.3. Environment-friendly and sustainable practices 1.4. Offering both affordable and high-quality products and services 1.5. Promotion and marketing strategies (e. g., on-line marketing)
2. Business operations	May include: 2.1 Purchasing 2.2 Accounting/Administrative work 2.3 Production/Operations/Sales
3. Internal controls	May include: 3.1 Accounting systems 3.2 Financial statements/reports 3.3 Cash management
4. Promotional/Advertising initiatives	May include: 4.1 Use of tarpaulins, brochures, and/or flyers 4.2 Sales, discounts and easy payment terms 4.3 Use of social media/Internet 4.4 "Service with a smile" 4.5 Extra attention to regular customers

EVIDENCE GUIDE

1. Critical aspects of competency	Assessment requires evidence that the candidate : 1.1 Demonstrated basic entrepreneurial skills 1.2 Demonstrated ability to conceptualize and plan a micro/small enterprise 1.3 Demonstrated ability to manage/operate a micro/small-scale business
2. Resource Implications	The following resources should be provided: 2.1 Simulated or actual workplace 2.2 Tools, materials and supplies needed to demonstrate the required tasks 2.3 References and manuals
3. Methods of Assessment	Competency in this unit may be assessed through : 3.1 Written examination 3.2 Demonstration/observation with oral questioning 3.3 Portfolio assessment with interview 3.4 Case problems
4. Context of Assessment	1.1 Competency may be assessed in workplace or in a simulated workplace setting 1.2 Assessment shall be observed while tasks are being undertaken whether individually or in-group

COMMON COMPETENCIES

UNIT OF COMPETENCY : VALIDATE VEHICLE SPECIFICATION

UNIT CODE : ALT723211

UNIT DESCRIPTOR : This unit covers the knowledge, skills and attitude to check body type of the vehicle, check vehicle engine type, check vehicle specifications and complete validation of vehicle specification.

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Check body type of the vehicle	1.1 <i>Kind of vehicle</i> is determined according to job order. 1.2 <i>Vehicle dimensions</i> is determined according to manual. 1.3 <i>Vehicle weight</i> is determined according to the manual. 1.4 <i>Body shape</i> is determined according to the manual. 1.6 <i>Power train</i> is determined according to the manual. 1.7 Safety practices are applied following OSHS	● Kind of vehicle - Aerodynamics - Vehicle Dynamics - Body shapes - Power train - Major dimensions ● Vehicle specifications - Vehicle performance - Weight & Measurements ● Automotive history ● Documentation/ Accomplishing checklist ● Resources information - Bulletin - Shop manual ● OSHS ● PPEs Attitude: ● Patience ● Attention to details	● Identifying kind of vehicle, dimensions, weight, body shape, and power train ● Accomplishing checklist ● Estimating visually dimensions and masses ● Utilizing resource information ● Wearing PPEs ● Applying safety practices
2. Check vehicle engine type	2.1 <i>Engine type</i> is identified according to industry standards. 2.2 Engine <i>fuel/energy system</i> is identified according to manual. 2.3 <i>Engine components</i> are	● Principles of internal combustions ● Principles of Electricity and motors ● History of engines ● Hybrid technology ● Resources information - Bulletin - Shop manual	● Identifying engine type, parts & components ● Identifying fuel systems or energy systems ● Utilizing resource information

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
	identified following manual.		
3. Check vehicle specifications	3.1 VIN plate is inspected for specification of vehicle according to manual. 3.2 Vehicle specification is verified according to vehicle reference materials. 3.3 Vehicle modifications and conversions are checked following the manual. 3.3 Vehicle conversions are inspected following the manual.	- Fundamentals of Automotive engineering: - Understanding of power & torque - Gear Ratios - Vehicle Regulations - Knowledge of vehicle performance - Knowledge in Vehicle manufacturing process - Knowledge of vehicle use - Automotive history - Knowledge in specifications - Reading of brochure, owner's manuals - Reading of Resources information - Bulletin - Shop manual	- Reading vehicle reference materials - Conducting vehicle inspection for modification and conversion - Comparing actual vehicle and specification sheets - Utilizing resource information
4. Complete validation of vehicle specification	4.1 Vehicle ownership is verified using repair order and vehicle reference materials. 4.2 Dealers check sheet is accomplished following industry standards. 4.3 Dealers check sheet is submitted to immediate superior following industry standards.	- Reporting to immediate superior - Documentation/ Accomplishing checklist Attitude: - Accuracy	- Verifying vehicle ownership - Accomplishing dealers check sheet - Reporting skills

RANGE OF VARIABLES

VARIABLE	RANGE
1. Kind of Vehicle	May include: 1.1 Motorized 1.2 Not Motorized 1.3 On-Road 1.4 Off-Road 1.5 Passenger 1.6 Commercial 1.7 Utility 1.8 Manned 1.9 Unmanned 1.10 Remote control 1.11 Automated/Self Driving 1.12 Guided
2. Vehicle Dimensions	May include: 2.1 Overall length 2.2 Overall width 2.3 Overall height 2.4 Wheelbase 2.5 Tread 2.6 Minimum running ground clearance 2.7 Room Length 2.8 Room Width 2.9 Room Height 2.10 Overhang front 2.11 Overhang rear 2.12 Angle of approach 2.13 Angle of departure
3. Vehicle Weight	May include: 3.1 Gross weight 3.2 Curb weight 3.3 Tare weight 3.4 Net weight
4. Body Shape	May include: 4.1 Sedan 4.2 Coupe 4.3 Hardtop 4.4 Convertible 4.5 Multipurpose vehicle (MPV) 4.6 Sports utility vehicle (SUV) 4.7 Truck 4.8 Tractor Head 4.9 Trailer 4.10 Special Utility Truck 4.11 Bus 4.12 Mini Bus 4.13 Articulated bus 4.14 Asian Utility Vehicle (AUV)

VARIABLE	RANGE
5. Power Train	May include: 5.1 Front Wheel Drive 5.2 Rear Wheel Drive 5.3 4x2 5.4 4x4 5.5 Limited Slip Differential (LSD) 5.6 Manual Transmission 5.7 Automatic Transmission 5.8 Continuously Variable Transmission
6. Engine Type	May include: 6.1 Internal Combustion Engine 6.2 Electric Motor
7. Fuel/Energy System	May include: 7.1 Diesel Fuel 7.2 Gasoline Fuel 7.3 Compressed Natural Gas (CNG) 7.4 Liquefied Petroleum Gas (LPG) 7.5 Methanol 7.6 Hydrogen 7.7 Biodiesel 7.8 Solar Cell 7.9 Fuel Cell
8. Engine Components	May include: 8.1 Intake System 8.2 Electrical System 8.3 Cooling System 8.4 Exhaust System 8.5 Valve Train System 8.6 Cylinder Head 8.7 Engine Block 8.8 Lubricating System
9. Vehicle reference materials	May include: 9.1 Warranty booklet 9.2 Brochure of the vehicle 9.3 Vehicle registration
10. Dealers check sheet	May include: 10.1 Vehicle mileage 10.2 Owner's information 10.3 Damage

EVIDENCE GUIDE

1. Critical Aspects of Competency	Assessment requires evidence that the candidate: 1.1 Checked body type of the vehicle 1.2 Checked vehicle engine type 1.3 Checked vehicle specifications 1.4 Completed validation of vehicle specification
2. Resource Implications	The following resources should be provided: 2.1 Workplace: Real or simulated work area 2.2 Appropriate vehicle or model equivalent 2.3 Materials relevant to the activity 2.4 Resource information, references, and manual
3. Method of Assessment	Competency in this unit may be assessed through: 3.1 Direct Observation 3.2 Interview 3.3 Third Party Report 3.4 Written exam 3.5 Demonstration with Oral questioning
4. Context of Assessment	4.1 Competency may be assessed individually in the actual workplace or through accredited institution

UNIT OF COMPETENCY : MOVE AND POSITION VEHICLE

UNIT CODE : ALT832212

UNIT DESCRIPTOR : This unit involves the skills and knowledge and attitudes required to move and position vehicle safely including systematic and efficient control of all vehicle functions.

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Prepare vehicle for operation	1.1 Vehicle multi point inspection is conducted according to industry practice. 1.2 Cockpit Drill is performed according to industry practice. 1.3 Vehicle is start-up following owner's manual. 1.4 Parking brake is engaged according to industry practice.	• Revolutions per minute during idle • Manual, automatic and CVT Transmission • Vehicle parts, components and functions • Inspection procedures • Owner's manual • Safety procedures	• Performing Cockpit Drill • Conducting Vehicle Multi point inspection • Starting the engine • Using owner's manual
2. Position vehicle	2.1 Workshop hazards are identified and avoided as per standard operating procedures. 2.2 Vehicle is moved according to Occupational Health and Safety Standards. 2.3 Workshop rules and regulations are recognized according to standard procedures.	• Revolutions per minute in running condition • Kilometer per hour • Estimation/ timing • Manual, automatic and CVT Transmission • Diesel, Gasoline and EV engines • Vehicle parts, components and functions • Defensive driving • Owner's Manual • Safety procedures	• Skills in positioning vehicle • Vehicle positioning estimation skill • Identifying workshop signs and markings
3. Park and stop the vehicle	3.1 Vehicle is positioned according to parking rules and regulations .	• Vehicle parts, components and functions • Inspection procedures • Owner's Manual	• Vehicle positioning estimation skills • Identifying parking signs and markings

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
	3.2 Parking brake is engaged according to industry practice. 3.3 Electrical devices are turned off based on manufacturer's specification. 3.4 Vehicle is shut-off following owner's manual	● Procedure in shutting-off vehicle ● Safety procedures ● Parking rules and regulations	

RANGE OF VARIABLES

VARIABLE	RANGE
1. Multi point inspection	May include: 1.1 Check for any obstruction 1.2 Check external condition 1.3 Check internal condition 1.3.1 Manual transmission 1.3.2 Automatic transmission 1.4 Check vehicle drivability
2. Cockpit Drill	May include: 2.1 Car mirror adjustments 2.3 Steering the car 2.4 How to change gears 2.5 Use of parking brake 2.6 Doors, Seat, Steering, Seat belt and Mirrors 2.7 Foot controls 2.8 Hand controls 2.9 Auxiliary controls (indicators)
3. Workshop hazards	May include: 3.1 Workshop tools and materials 3.2 Workshop equipment 3.3 Other vehicles 3.4 Other people 3.5 Oil spills 3.6 Loose parts
4. Parking rules and regulation	May include: 4.1 Parallel parking 4.2 Horizontal parking 4.3 Park facing the wall
5. Electrical devices	May include: 5.1 Lights 5.2 Air conditioning 5.3 Wiper 5.4 Radio

EVIDENCE GUIDE

1. Critical aspects of competency	Assessment requires evidence that the candidate: 1.1 Prepared vehicle for operation 1.2 Positioned the vehicle 1.3 Parked and stopped the vehicle 1.4 Used owner's manual
2. Resource implication	The following resources MUST be provided: 2.1 Workshop range/area 2.2 Service working bay 2.3 Appropriate vehicle for moving and positioning 2.4 Owner's manual
3. Method of assessment	Competency MUST be assessed through: 3.1 Demonstration with oral questioning 3.2 Written exam 3.3 Interview 3.3 Direct observation
4. Context of assessment	4.1 Competency may be assessed individually in the actual workplace or through accredited institution

UNIT OF COMPETENCY : UTILIZE AUTOMOTIVE TOOLS

UNIT CODE : ALT723213

UNIT DESCRIPTOR : This unit covers the knowledge and skills in selecting and using automotive power tools, hand tools and tool keeping.

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Prepare automotive tools	1.1 Automotive tools are identified according to their classification and specification. 1.2 Automotive tools and attachments are selected according to job requirements. 1.3 Automotive tools and attachments are inspected for defects and damages according to manufacturers and work place procedures. 1.4 Safety practices are applied following OSHS.	• Understanding power to size ratio • Leverage • Types of power tools and hand tools • Uses of automotive power tools and hand tools • Defects and damages of automotive tools and attachments • Handling of tools • Interpretation of contents of users manuals • Safety procedures • Wearing of PPE	• Identifying defects or damages of tools before use • Knowledgeable in proper handling of tools • Identifying tools required for the job • Inspecting the area where power tools will be used.
2. Use automotive tools	2.1 Attachments are mounted to automotive tools according to job requirements 2.2 Power tools are connected to power sources according to operation's manual 2.3 Power tools are operated according to operation's manual 2.4 Hand tools are utilized according to operation's manual	• Use of automotive tools • Application of Torque and pressure • Unit conversion of torque • English and metric system • Types of hand tools • Types of power tools • Fundamentals of automotive hand tools and power tools	• Analytical skills • Technical literacy • Mounting attachments to automotive tools • Connecting power tools to power sources • Operating power tools • Utilizing hand tools • Wearing PPEs • Applying safety practices • Following manuals

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
	2.5 PPEs are worn in accordance to OSHS.	• Interpretation of contents of users manuals • OSHS • Resources information - Bulletin - Shop manual	
3. Maintain automotive tools	3.1 Automotive tools and attachments are cleaned according to user's manual. 3.2 Automotive tools and attachments are checked for serviceability according to workplace and manufacturers procedures. 3.3 Defects and damages are reported to immediate superior following industry standards. 3.4 Automotive tools and attachments are stored according to workplace procedures. 3.5 Safety practices are applied following OSHS. 3.6 Wastes are disposed following environmental law and regulations.	• Different types of power tools and hand tools • Techniques in tool Arrangement • Fundamentals of automotive tools • Cleaning of automotive tools • Labeling and arranging of power tools and hand tools • Safety practices • Procedures in maintaining of power tools and hand tools • Tagging of damaged/worn power tools and hand tools • Reporting damage power tools and hand tools • Proper disposal of damaged tools • Proper disposal of chemicals used for cleaning • OSHS • Environmental law and regulations • 5S of good housekeeping • 3Rs	• Sorting of tools • Skills in creating reports • Cleaning of tools • Checking, cleaning and storing automotive tools and attachments • Reporting defects and damages • Disposing wastes • Practicing safety procedures

RANGE OF VARIABLES

VARIABLE	RANGE
1. Automotive tools	May include: 1.1 Power tools 1.1.1 Electric power tools 1.1.1.1 Electric drill 1.1.2 Pneumatic tools 1.2 Basic tools 1.3 Special service tools (SST)
2. Power sources	May include: 2.1 Electric source 2.2 Pneumatic or air 2.3 Hydraulic
3. Basic tools	May include: 3.1 Wrenches 3.2 Pliers 3.3 Screw drivers 3.4 Power handle 3.5 Ratchet 3.6 Multi-meter tester 3.7 Flash light 3.8 Rubber mallet 3.9 Hammer 3.10 Jack 3.11 Jack stand 3.12 Choke
4. Attachments	May include: 4.1 Bits 4.2 Sockets 4.3 Extension
5. Defects and damages	May include: 5.1 Tools 5.1.1 Cracks 5.1.2 Breakage 5.1.3 Deformity 5.1.4 Looseness 5.1.5 Corrosions 5.1.6 Leaks 5.2 Attachments 5.2.1 Cracks 5.2.2 Breakage 5.2.3 Deformity 5.2.4 Looseness 5.2.5 Corrosions
6. Personal protective equipment (PPEs)	May include: 6.1 Goggles 6.2 Gloves 6.3 Hard hat 6.4 Safety shoes

VARIABLE	RANGE
	6.5 Dust mask
7. Wastes	May include: 7.1 Dead batteries 7.2 Deformed, cracked, broken bits/sockets/extensions 7.3 Used cleaning chemicals 7.4 Used oil 7.5 Contaminated cleaning materials

EVIDENCE GUIDE

1. Critical aspects of competency	Assessment require evidence that the candidate understands the applications and guidelines specified by the manufacturer. 1.1 Prepared automotive tools 1.2 Used Power tools 1.3 Used Hand tools 1.4 Maintained and stored automotive tools 1.5 Disposed wastes 1.6 Applied safety measures
2. Resource implication	The following resource MUST be provided. 2.1 Appropriate power tools and hand tools 2.2 Tools and materials relevant for training 2.3 Proper place for storage and disposal 2.4 Work shop manuals
3. Method of assessment	Competency MUST be assessed through. 3.1 Written examination 3.2 Demonstrations with oral questioning 3.3 Direct observation 3.4 Third party report 3.5 Interview
4. Context of assessment	4.1 Competency may be assessed individually in the actual workplace or through accredited institution

UNIT OF COMPETENCY : PERFORM MENSURATION AND CALCULATION

UNIT CODE : ALT723214

UNIT DESCRIPTOR : This unit covers the knowledge and skills on how to use automotive measuring tools.

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Select measuring instruments	1.1 Component to be measured is identified based on job requirements. 1.2 Automotive measuring instrument is identified based on job requirements. 1.3 Correct specifications are obtained from repair manual. 1.4 Measuring tools are calibrated in line with job requirements. 1.5 Measuring instruments are checked for accuracy and adjusted according to manufacturer's manual 1.6 Defective measuring instruments are reported and returned to toolkeeper following industry standards 1.7 Safety practices are applied following OSHS	• Category of measuring instruments • Types and uses of measuring instruments • Shapes and Dimensions • Use of user's manual • Workshop procedures in reporting defective instruments • Characteristics of defective measuring instruments • Procedure in preparing report • OSHS in calibrating measuring instruments • Calibration of measuring tools • Inspection of measuring tools • Segregation and reporting of defective measuring instruments	• Identifying and selecting measuring instruments • Visualizing objects and shapes • Calibration skills • Identifying defective measuring instruments • Reporting skills • Applying safety practices • Obtaining correct specifications • Checking measuring instruments for accuracy • Reporting and segregating defective measuring instruments
2. Carry out measurements and calculation	2.1 Automotive measuring instrument is selected to achieve required outcome in line with job requirements	• Formulas for volume, areas, perimeters of plane and geometric figures	• Performing calculation • Applying formulas for volume, areas, perimeters of plane and geometric figures

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
	2.2 Accurate measurements are obtained in line with job requirements 2.3 Calculation needed to complete work tasks are performed using mathematical operations. 2.4 Numerical computation is self-checked and corrected for accuracy following manufacturer's workshop manual 2.3 Tools' limit of accuracy are read following manufacturer's workshop manual 2.4 Report is submitted to immediate supervisor following industry standard operating procedure 2.5 Safety practices are applied following OSHS	• Different automotive measuring instruments • Calculation & measurement • Four fundamental operation • Linear measurement • Dimensions • Unit conversion • Ratio and proportion • Handling of measuring instruments • Tools' limit of accuracy • OSHS • PPEs	• Handling measuring instruments • Selecting automotive measuring instruments • Obtaining accurate measurements • Performing calculation • Self-checking and correcting numerical computation • Reading tools' limit of accuracy • Applying OSHS • Wearing of PPEs
3. Maintain measuring instruments	3.1 Measuring instruments are handled following manufacturer's manual 3.2 Measuring instruments are cleaned following manufacturer's manual. 3.3 Instruments are stored according to manufacturer's specifications and standard operating procedures. 3.4 Safety practices are applied	• Types of measuring instruments and their uses • Safe handling procedures in using measuring instruments • Four fundamental operation of mathematics • Formula for volume, area, perimeter and other geometric figures • 5S of good housekeeping • Waste management	• Handling and maintaining measuring instruments • Disposing wastes • Practicing good housekeeping • Applying safety practices

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
		• Storing of measuring instruments • OSHS	

RANGE OF VARIABLES

VARIABLE	RANGE
1. Automotive measuring instruments	May include: 1.1 Torque wrench 1.2 Vernier caliper 1.3 Micrometer (inside and outside) 1.4 Dial gauge 1.5 Feeler gauge 1.7 Pitch/thread gauge 1.8 Multi-tester (analog/digital) 1.9 Vacuum Gauge 1.10 Tire depth gauge 1.11 Battery tester 1.12 Steel tape 1.13 Ruler
2. Calculation	May include: 2.1 Volume 2.2 Area 2.3 Displacement 2.4 Inside diameter 2.5 Circumference 2.6 Length 2.7 Thickness 2.8 Outside diameter 2.9 Taper 2.10 Out of roundness 2.11 Voltage 2.12 Resistance 2.13 Current 2.14 Pressure 2.15 Clearance 2.16 Distortion/run-out 2.17 Torque conversion 2.18 Temperature
3. Mathematical operations	Includes: 3.1 Addition 3.2 Subtraction 3.3 Multiplication 3.4 Division 3.5 Fractions 3.6 Percentages 3.7 Mixed numbers

EVIDENCE GUIDE

1. Critical aspects of competency	Assessment requires evidence that the candidate perform the following: 1.1 Selected measuring instruments 1.2 Performed measurements and calculation 1.3 Maintained measuring instruments 1.4 Applied safety practices
2. Resource implications	The following resources MUST be provided: 2.1 Workplace: Real or simulated work area 2.2 Appropriate Automotive Measuring Tools & equipment 2.3 Materials relevant to the activity 2.4 Training vehicle or simulators 2.5 User's manual 2.6 Repair manual
3. Method of assessment	Competency MUST be assessed through: 3.1 Written exam 3.2 Demonstration with oral questioning 3.3 Third party report 3.4 Interview
4. Context of assessment	4.1 Competency may be assessed individually in the actual workplace or through accredited institution

UNIT OF COMPETENCY : UTILIZE WORKSHOP FACILITIES AND EQUIPMENT

UNIT CODE : ALT723215

UNIT DESCRIPTOR : This unit deals with inspecting and cleaning of work area including tools, equipment and facilities.
Storage of equipment, including operating of basic workshop equipment.

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Perform pre-operation activities	1.1 Workshop facilities are prepared according to work requirements. 1.2 Equipment are prepared according to work requirements. 1.3 Equipment are calibrated following users' manual. 1.4 Minor repairs are carried out based on users' manual. 1.5 Defective equipment are reported to immediate supervisor following company procedures. 1.6 Safety practices are applied following OSHS.	• Different areas of an automotive service facilities. • Preparation procedures of automotive service facilities • Different equipment in the automotive service facilities • Preparation procedures of automotive equipment • Minor repairs of automotive equipment • Report of defective equipment • Reporting procedures for defective equipment • OSHS practices related to the preparation of facilities and equipment • Workshop facilities and equipment	• Preparing work area • Preparing equipment • Calibrating equipment • Repairing minor equipment issues • Reporting defective equipment • Applying safety practice • Following manuals

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
2. Use facilities and equipment	2.1 Equipment is operated according to operation <i>manual</i>. 2.2 Facilities are utilized according to workshop procedures. 2.3 Equipment performance is monitored following users' <i>manual</i>. 2.4 Facilities functionalities are monitored following workplace procedures. 2.5 Safety practices are applied following OSHS.	• Operate Equipment • Identify facilities required for task • Evaluate equipment operation • Inspect facility functionalities • OSHS practices related to operation of facilities and equipment • Manuals in utilizing facility and equipment • Monitoring procedure of equipment's performance • Evaluate equipment operation • Inspection of facility functionalities	• Operating equipment • Utilizing facility • Monitoring equipment performance • Monitoring functionalities of facility • Practicing safety • Following manual
3. Conduct post-operation activities	3.1 Workshop facilities are restored according to 5S of good housekeeping. 3.2 <i>Equipment</i> are cleaned and stored according to good housekeeping. 3.3 Wastes are disposed following waste management procedure and OSHS. 3.4 <i>PPEs</i> and Safety practices are applied following OSHS. 3.5 Report is prepared based on workshop procedure.	• 5S of Good housekeeping • 3Rs/ Waste segregation and disposal • Restoration of the facilities • Maintenance and storage of Equipment • OSHS • Preparation of report	• Restoring workshop facilities properly • Cleaning Equipment • Storing equipment in proper location • Disposing waste materials • Reporting facilities and equipment condition • Practicing safety • Practicing 5S and 3Rs

RANGE OF VARIABLES

VARIABLE	RANGE
1. Equipment	May include: 1.1 Lifter (Two Post Lifter / Four Post Lifter/ Scissor type) 1.2 Crocodile Jack 1.3 Jack Stand 1.4 Air Compressor 1.5 Oil drain
2. Workshop facilities	May include: 2.1 Service Stall / Working Bay / Workshop areas for servicing/repairing light and/or heavy vehicle and/or plant transmissions and/or outdoor power equipment 2.2 Overhauling Room 2.3 Electrical / Air-con Room 2.4 Inspection Area 2.5 Open workshop/garage and enclosed, ventilated office area 2.6 Car wash area 2.7 Other variables may include workshop with: 2.7.1 Mess hall 2.7.2 Wash room 2.7.3 Comfort room 2.7.4 Storage Room 2.7.5 Training Room
3. Manuals	May include: 3.1 Vehicle/plant manufacturer specifications 3.2 Company operating procedures 3.3 Industry/Workplace Codes of Practice 3.4 Product manufacturer specifications 3.5 Industry Occupational Health & Safety 3.6 Equipment Operation Guidelines 3.7 Service/workshop/repair manual
4. PPEs	May include: 4.1 Gloves 4.2 Apron 4.3 Goggles 4.4 Safety shoes 4.5 Uniforms 4.6 Cap 4.7 Safety helmet
5. Minor repairs	May include: 3.1 Lubrication 3.2 Bolt tightening 3.3 Worn-out parts replacement

EVIDENCE GUIDE

1. Critical aspects of competency	Assessment requires evidence that the candidate: 1.1 Performed pre-operation activities 1.2 Used facilities and equipment 1.3 Conducted post-operation activities 1.4 Applied safety practices and good housekeeping 1.5 Disposed wastes
2. Resource implications	The following resources should be provided: 2.1 Workplace: Real or simulated work area 2.2 Appropriate Equipment 2.3 Materials relevant to the activity 2.4 Manuals/references 2.5 PPEs 2.6 Fire Extinguishers
3. Method of assessment	Competency in this unit may be assessed through: 3.1 Written exam 3.2 Demonstration with oral questioning 3.3 Direct observation
4. Context of assessment	4.1 Competency may be assessed individually in the actual workplace or through accredited institution

UNIT OF COMPETENCY : PREPARE SERVICING PARTS AND CONSUMABLES

UNIT CODE : ALT723216

UNIT DESCRIPTOR : This unit of competency covers the ability to prepare parts and consumables for gasoline and diesel engines in conducting preventive maintenance.

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Identify parts and consumables	1.1 <i>Parts and consumables</i> are <i>determined</i> according to job requirements. 1.2 Availability of parts and consumables are confirmed based on stock. 1.3 <i>Indirect materials</i> are identified according to job requirements. 1.4 <i>Hazardous parts and consumables</i> are identified according International standards. 1.5 Safety practices are applied according to OSHS.	• Job requirements • Safety practices • Understanding manuals • Hazardous parts and consumables • Solid waste management act (RA 6969) • Wearing of PPE's • OSHS • Proper storage of materials • Chemical contents of consumables • Composition of consumables • Quality of parts and consumables • Computation for quantity of parts and consumables • Vehicle specifications • Identifying Part no. • Awareness in part number • Updated type of parts and consumables	• Determining parts and consumables • Reading and interpreting job requirements • Identifying required parts & consumables • Understanding safety practices • Determining quantity and quality of parts and consumables • Confirming availability of parts and consumables • Identifying indirect materials • Identifying hazardous parts and consumables • Applying safety practices • Understanding safety practices • Following manuals
2. Retrieve and withdraw parts and consumables	2.1 Requisition slip is prepared according to identified parts and consumables. 2.2 Withdrawal of parts and materials are recorded. 2.3 Quantity of parts and consumables	• Job requirements • Safety practices • Understanding manuals • Hazardous parts and consumables • Solid waste management act (RA 6969)	• Reading and interpreting requisition slip • Validating quantity of parts and materials • Handling parts and consumables

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
	are validated according to job requirements 2.4 Parts and materials are handled following safety procedures.	● Wearing of PPE's ● Updated types of parts & consumables for proper usage	
3. Complete work process	3.1 Used parts and consumables are labeled and segregated 3.2 Used parts are packed and returned to customers 3.3 Consumables are collected for recycling 3.4 PPEs are worn following OSHS. 3.5 Wastes are disposed according to RA 6969.	● Labeling and segregation of used parts and consumables ● Job requirements ● Safety practices ● 3Rs ● Solid waste management act (RA 6969) ● Wearing of PPE's	● Waste segregation and disposal of parts & consumables according to RA 6969

RANGE OF VARIABLES

VARIABLE	RANGE
1. Parts and consumables	May include: 1.1 Engine oil 1.2 Clutch fluid 1.3 Transmission oil 1.4 Differential oil 1.5 Power steering fluid 1.6 Brake fluid 1.7 Engine coolant 1.8 Engine oil filter 1.9 Fuel filter 1.10 Air cleaner element 1.11 Feed pump strainer 1.12 Sparkplugs (Gasoline engine) 1.13 Battery 1.14 Air cleaner 1.15 Tire 1.16 Wiper blade 1.17 A/C pollen filter 1.18 Bulb 1.19 Brake pad/brake shoe 1.20 Clutch lining
2. Determining parts and consumables	May include: 2.1 Quantity 2.2 Quality
3. Indirect materials	May include: 3.1 Rags 3.2 Saw dust 3.3 Cleaning fluids 3.4 Sand paper
4. Hazardous parts consumables	May include: 4.1 Batteries 4.2 Used oil 4.3 Used fluids 4.4 Used coolant 4.5 Used parts 4.6 Used oil filter
5. Wastes	May include: 5.1 Contaminated consumables 5.2 Contaminated parts

EVIDENCE GUIDE

1. Critical aspects of competency	Assessment requires evidence that the candidate: 1.1 Identified parts and consumables 1.2 Retrieved and withdrawn parts and consumables 1.3 Completed work process 1.4 Applied safety practices
2. Resource implications	The following resources should be provided: 2.1 Workplace: Real or simulated work area 2.2 Materials relevant to the activity 2.3 Repair manuals and related reference materials
3. Method of assessment	Competency in this unit may be assessed through: 3.1 Direct observation 3.2 Interview 3.3 Written examination 3.4 Demonstration with oral questioning 3.5 Third party report
4. Context of Assessment	4.1 Competency may be assessed individually in the actual workplace or through accredited institution

UNIT OF COMPETENCY : PREPARE VEHICLE FOR SERVICING AND RELEASING

UNIT CODE : ALT723217

UNIT DESCRIPTOR : This unit covers the knowledge, skills, and attitudes needed in identifying and preparing the vehicle for servicing and releasing.

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Receive vehicle	1.1 Vehicle is located following company standard. 1.2 Checklist is validated for exterior and interior items in accordance with vehicle checklist . 1.3 Job Order is checked for proper assignment according to work classification . 1.4 Work bay for vehicle is designated based from Job Order. 1.5 Vehicle is moved on the designated work bay .	● Identification of basic vehicle components ● Types of defects ● Read & understand Job Order ● Flat rate time ● Use of PPEs ● Adherence to safety procedures ● Vehicle checklist ● Work classification ● Work bay Attitudes: ● Patient ● Attention to details ● Honest ● Time Conscious	● Completing vehicle checklist ● Classifying work to be performed ● Assigning work bay ● Validating checklist for exterior and interior items ● Checking job order for proper assignment ● Identifying vehicle ● Moving vehicle to designated work bay
2. Prepare vehicle for servicing	2.1 Protective covers are installed prior to servicing based on workshop operating standards 2.2 Vehicle is positioned and set-up for lifting according to repair order. 2.3 Vehicle is lifted for servicing following manufacturer's manual. 2.4 Safety practices are applied following safety procedures.	● Familiarization on equipment & facilities ● Time estimation of completion ● Vehicle tagging ● Types of protective covers ● Setting-up of vehicle for lifting ● Read & understand repair order ● Use of PPEs ● Use of safety gears ● OSHS ● Adherence to safety procedures	● Understanding of vehicle status ● Installation of protective covers ● Positioning vehicle ● Operating lifter ● Moving vehicle ● Setting-up vehicle for lifting ● Practicing safety

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
		Attitudes: ● Patient ● Attention to details ● Honest ● Time Conscious	
3. Prepare vehicle for releasing	3.1 Job done is confirmed according to repair order. 3.2 Quality check is done based from repair order. 3.3 Transfer of vehicle to wash bay is coordinated according to SOP. 3.3 Vehicle is endorsed to quality control person following workplace procedure.	● Familiarization of equipment & facilities ● Read & understand repair order ● Confirmation of job done ● Quality standards checking ● Coordination of transferring vehicle ● Endorsement procedures for vehicle Attitudes: ● Patient ● Attention to details ● Honest ● Time Conscious	● Confirming job done ● Performing quality checking ● Coordinating transfer of vehicle to wash bay ● Endorsing and turning-over vehicle

RANGE OF VARIABLES

VARIABLE	RANGE
1. Vehicle checklist	May include: 1.1 External scratches, accessories, items, dents, damages and cracks 1.2 Internal items, scratches, noticeable damages, including spare tire, tools, and loose items 1.3 Standard items that are not present during inspection 1.4 Valuable/personal belongings
2. Work classification	May include: 2.1 Body and Paint repair 2.2 General Job repair 2.3 Periodic maintenance service (PMS)
3. Work bay	May include: 3.1 Service Stall / Working Bay / Workshop areas for servicing/repairing light and/or heavy vehicle and/or plant transmissions and/or outdoor power equipment 3.2 Overhauling Room 3.3 Electrical / Air-con Room 3.4 Inspection Area 3.5 Open workshop/garage and enclosed, ventilated office area
5. Protective covers	May include but not limited to: 5.1 Seat Cover 5.2 Steering Wheel Cover 5.3 Handbrake Cover 5.4 Shift Knob Cover 5.5 Fender Cover 5.6 Paper mat

EVIDENCE GUIDE

1. Critical aspects of competency	Assessment requires evidence that the candidate: 1.1 Received vehicle 1.2 Prepared vehicle for servicing 1.3 Prepared vehicle for releasing 1.4 Applied safety practices
2. Resource implications	The following resources MUST be provided: 2.1 Workplace: Real or simulated work area 2.2 Appropriate Tools & Equipment 2.3 Materials relevant to the activity 2.4 Manuals and references
3. Method of assessment	Competency may be assessed through: 3.1 Direct observation 3.2 Demonstration with Oral questioning 3.3 Interview 3.4 Written Evaluation 3.5 Third Party Report
4. Context of assessment	4.1 Competency may be assessed individually in the actual workplace or through accredited institution

CORE COMPETENCIES

UNIT OF COMPETENCY : DIAGNOSE AND REPAIR ELECTRONICALLY CONTROLLED TRANSMISSION

UNIT CODE : ALT723321

UNIT DESCRIPTOR : This unit identifies the competence required to diagnose and repair faults in the electronically controlled transmission. It involves preparing for the task, selecting the correct diagnostic procedure, carrying out the diagnosis and the repair, performing post repair testing, and completing workplace processes and documentation.

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Prepare to diagnose electronically controlled transmission	1.1 Safety practices are applied following OSHS. 1.2 Job requirements are determined from workplace instructions. 1.3 Diagnostic information is sourced and interpreted according to manufacturer specification and workplace procedures. 1.4 Hazards associated with the work are identified following industry criteria . 1.5 Risks are managed based on industry criteria . 1.6 Tools and equipment are selected and checked for serviceability according to industry criteria . 1.7 Defective tools and equipment	1.1 Occupational Safety and Health Standard (OSHS) 1.2 Waste management 1.3 Sourcing out and interpretation of diagnostic and dismantling information 1.4 Diagnostic symptoms and troubleshooting guide 1.5 Disassembly procedure 1.6 Service/Repair manual 1.7 Tools, equipment and materials in diagnosing and repair 1.8 Interpretation of job requirements 1.9 Serviceability of tools and equipment	1.1 Interpreting job requirements from workplace instructions 1.2 Clarifying instructions 1.3 Locating appropriate sources of information 1.4 Analyzing diagnostic symptoms and disassembly options 1.5 Selecting and checking tools and equipment 1.6 Reporting Defective tools and equipment 1.7 Checking and reporting the availability of materials 1.8 Applying safety practices

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
	are reported following workplace procedures. 1.8 Parts and supplies are sourced following workplace procedures.		
2. Perform diagnosis	2.1 Diagnostic tests are carried out according to industry criteria 2.2 Diagnostic tools are operated following manufacturer's specification. 2.3 Faults and its causes are determined from diagnostic test results. 2.4 Diagnosis findings and recommendations are reported according to industry criteria . 2.5 Safety practices are applied following Occupational Health and Safety Standard (OSHS).	2.1 Diagnostic symptoms and troubleshooting guide 2.2 Identification, function and operation of electronically controlled transmission 2.3 Mensuration 2.4 Arithmetic operations 2.5 Use of measuring devices 2.6 Reporting procedures 2.7 Oil condition and viscosity 2.8 Noise 2.9 OSHS 2.10 Wearing of PPEs 2.11 Different faults of electronically controlled transmission 2.12 Operating principles of electronically controlled transmission 2.13 Diagnostic testing procedures for electronically controlled transmission 2.14 Industry criteria	2.1 Analyzing diagnostic symptoms 2.2 Interpreting information from manufacturer and workshop literature 2.3 Performing diagnostic tests 2.4 Inspecting electronically controlled transmission 2.5 Comparing inspection results 2.6 Identifying faults and its causes 2.7 Reporting and recommendations findings 2.8 Applying safety practices 2.9 Mensuration skills 2.10 Applying arithmetic operations 2.11 Reporting diagnostic findings and make repair recommendations

3. Repair electronically controlled transmission components	3.1 Repair information is sourced and interpreted following industry criteria. 3.2 Repair options and solutions are analyzed and selected based on manufacturer's manual circumstances at hand. 3.3 Repair is carried out according to manufacturer specification workshop procedure and industry criteria. 3.4 Functionality test is carried out to confirm fault rectification according to workplace procedures. 3.5 Rectification is done based on the result of functionality test. 3.6 Documentation report is prepared based on analysis result.	3.1 Types, characteristics and operating principles of electronically controlled transmission 3.2 Electronically controlled transmission dismantling procedures 3.3 Electronically controlled transmission component inspection, measuring and evaluation procedures 3.4 Component repair method 3.5 Mensuration 3.6 Arithmetic operation 3.7 Special service tools (SST) 3.8 Sourcing of electronically controlled transmission components 3.9 OSHS 3.10 Wearing of PPEs 3.11 Replacement of electronically controlled transmission components 3.12 3Rs 3.13 5S of Good Housekeeping 3.14 Tolerances and clearances of components of electronically controlled transmission 3.15 Industry criteria 3.16 Assembly procedures 3.17 Post-assembly testing	3.1 Interpreting information from manufacturer and workshop literature 3.2 Measuring electronically controlled transmission components 3.3 Using basic mathematical operations, including addition, subtraction, multiplication and division 3.4 Pull down skills 3.5 Arithmetic skills 3.6 Mensuration skills 3.7 Dismantling electronically controlled transmission 3.8 Cleaning and evaluating electronically controlled transmission components 3.9 Sourcing standard components 3.10 Using special service tools (SST) 3.11 Operating equipment 3.12 Replacing out of standard components 3.13 Assembling electronically controlled transmission 3.14 Measuring tolerances and clearances of components of electronically controlled transmission 3.15 Completing assembly of electronically
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ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
			controlled transmission 3.16 Carrying out post-assembly testing 3.17 Applying safety practices
4. Complete work processes	4.1 Final inspection is conducted based on workplace procedure. 4.2 Vehicle is presented to client according to industry practice. 4.3 Work area is restored following 5S of Good Housekeeping. 4.4 Wastes are managed following environmental rules and regulations. 4.5 Tools and equipment are checked and stored according to workplace procedures. 4.6 Workplace documentation is processed according to workplace.	4.1 Final inspection procedure 4.2 Turn-over of vehicle 4.3 Accomplishment of repair order and other forms 4.4 Job done 4.5 OSHS 4.6 Wearing of PPEs 4.7 5S of Good Housekeeping 4.8 Waste management 4.9 Checking and storage of tools and equipment 4.10 Workplace documents	4.1 Filling-out workplace documentation. 4.2 Reporting diagnostic findings and make repair recommendations 4.3 Conducting final inspection 4.4 Performing vehicle turn-over 4.5 Restoring work area 4.6 Managing wastes 4.7 Checking and storing tools and equipment 4.8 Wearing of PPEs 4.9 Applying safety practices 4.10 Accomplishing workplace documents

RANGE OF VARIABLES

VARIABLE	RANGE
1. Job requirements	Job requirements may include: 1.1 Perform mechanical system test 1.2 Removal/installation of shift mechanism 1.3 Inspection of selector lever 1.4 Inspection of shift lock system 1.5 Inspection of neutral start switch 1.6 Inspection of electronically controlled transmission fluid 1.7 Replacement/adjustment of electronically controlled transmission 1.8 Inspection of “M” position switch 1.9 Removal/inspection of “M” position switch 1.10 Inspection of Up switch 1.11 Removal/installation of Up switch 1.12 Inspection of Down switch 1.13 Removal/installation of Down switch 1.14 Removal/installation of coupler components 1.15 Removal/installation of oil pressure switch 1.16 Removal/installation of ON/OFF solenoid 1.17 Removal/installation of electric AT oil pump 1.18 Removal/installation of electronically controlled transmission 1.19 Removal/installation of drive plate 1.20 Replacement of differential oil seal 1.21 Replacement of control valve body oil seal 1.22 Removal/installation of oil cooler 1.23 Removal installation of control valve body 1.24 Perform initial learning 1.25 Configuration of electronically controlled transmission module 1.26 Diagnose available data trouble code 1.27 Replacement of electronically controlled transmission module 1.28 Replacement of power control module
2. Industry criteria	Industry criteria may include: 2.1 Manufacturer specifications 2.2 Repair manual 2.3 Workplace procedures 2.4 Safety and environmental requirements 2.5 Service history
3. Tools and equipment	Tools and equipment may include: 3.1 Tools: 3.1.1 Standard technician hand tools 3.1.2 Torque wrench 3.1.3 Wheel chocks 3.1.4 Plastic clip remover 3.1.5 Multi-meter tester

VARIABLE	RANGE
	3.1.6 Oil pressure gauge set with adapters 3.1.7 Engine lifting brackets (SST) 3.1.8 Engine support set (SST) 3.1.9 Support hanger (SST) 3.1.10 Ring gear brake set (SST) 3.1.11 Oil seal installer (SST) 3.1.12 Dust boot installer (SST) 3.1.13 Diagnostic scan tools 3.2 Equipment: 3.2.1 Lifter 3.2.2 Air reel 3.2.3 Workshop compressor with air line 3.2.4 Electrical reel 3.2.5 Oil drain tank 3.2.6 Transmission jack 3.2.7 Diagnostic scan tool
4. Parts and supplies	Parts and supplies may include: 4.1 Parts 4.1.1 Automatic transmission kit 4.2 Supplies 4.2.1 Rags 4.2.2 Electronically controlled transmission fluid 4.2.3 Penetrating oil 4.2.4 Silicon sealant 4.2.5 Shock absorbing foam 4.2.6 PPEs 4.2.7 Petroleum jelly 4.2.8 Onion skin paper
5. Diagnostic tests	Diagnostic tests may include: 5.1 Checking of oil condition and viscosity 5.2 Checking of voltage, continuity or resistance according to the standard 5.3 Checking of functionality and drivability 5.4 Checking of possible oil leaks
6. Diagnostic tools	Diagnostic tools may include: 6.1 Multi-meter tester 6.2 Oil pressure gauge set with adapters 6.3 Diagnostic scan tools
7. Repair options	Repair options may include: 7.1 Cleaning 7.2 Evaluation by measurement 7.3 Evaluation by comparison 7.4 Evaluation by visual
8. Repair	Repair may include: 8.1 Adjustment of inhibitor switch and shift lever 8.2 Replacement of inhibitor switch, cable, lever, and shift solenoid 8.3 Removal and installation of electronically controlled transmission

VARIABLE	RANGE
9. Functionality test	Functionality test may include: 9.1 Inspection of oil leak 9.2 Inspection of functionality and drivability
10. Documentation report	Documentation report includes: 10.1 Accomplishment report 10.2 Recommendation report
11. Workplace documentation	Workplace documentation may include: 11.1 Repair order 11.2 Inspection form 11.3 Diagnostic form 11.4 Service history

EVIDENCE GUIDE

1. Critical aspects of competency	Assessment requires evidence that the candidate: 1.1 Prepared to diagnose electronically controlled transmission. 1.1.1 Determined job requirements. 1.1.2 Selected and checked tools and equipment. 1.1.3 Applied safety practices. 1.2 Performed diagnosis. 1.2.1 Operated diagnostic tools. 1.2.2 Reported inspection findings and recommendations. 1.2.3 Applied safety practices. 1.3 Repaired electronic fuel injection system. 1.3.1 Carried out repairs. 1.3.2 Carried out functionality test. 1.3.3 Applied safety practices. 1.4 Completed work processes. 1.4.1 Conducted final inspection. 1.4.2 Accomplished workplace documentation.
2. Resource implications	The following resources MUST be provided: 2.1 Actual or simulated work place 2.2 Appropriate tools & equipment 2.3 Materials relevant to the activity 2.4 Manufacturer's repair manual 2.5 PPEs 2.6 Training vehicle 2.7 First aid kit
3. Method of assessment	Competency in this unit may be assessed through: 3.1 Demonstration/direct observation with oral questioning 3.2 Written exam 3.3 Portfolio
4. Context for assessment	4.1 Competency may be assessed individually in the actual workplace or simulation environment in TESDA accredited institutions

UNIT OF COMPETENCY : DIAGNOSE AND REPLACE CONTINUOUSLY VARIABLE TRANSMISSION

UNIT CODE : ALT723322

UNIT DESCRIPTOR : This unit identifies the competence required to diagnose and repair faults in the continuously variable transmissions. It involves preparing for the task, selecting the correct diagnostic procedure, carrying out the diagnosis and the repair, performing post-repair testing, and completing workplace processes and documentation.

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Prepare to diagnose continuously variable transmission	1.1 Job requirements are determined from workplace instructions. 1.2 Diagnostic information is sourced and interpreted according to manufacturer specification and workplace procedures. 1.3 Hazards associated with the work are identified based on industry criteria. 1.4 Risks are managed based on industry criteria. 1.5 Tools and equipment are selected and checked for serviceability according to industry criteria . 1.6 Defective tools and equipment are reported following workplace procedures. 1.7 Supplies are sourced following	1.1 OSHS 1.2 Waste management 1.3 Sourcing out and interpretation of diagnostic and dismantling information 1.4 Diagnostic symptoms and troubleshooting guide 1.5 Service manual 1.6 Tools, equipment and materials in diagnosing 1.7 Interpretation of job requirements 1.8 Serviceability of tools and equipment 1.9 Identification, function and operation of continuously variable transmission 1.10 Operating principles of continuously variable transmission	1.1 Interpreting job requirements from workplace instructions 1.2 Clarifying instructions 1.3 Locating appropriate sources of information 1.4 Analyzing diagnostic symptoms 1.5 Selecting and checking tools and equipment 1.6 Reporting defective tools and equipment 1.7 Checking and reporting the availability of materials 1.8 Applying safety practices

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
	workplace procedures. 1.8 Safety practices are applied following OSHS.		
2. Perform diagnosis	2.1 <i>Diagnostic tests</i> are carried out according to <i>industry criteria</i> . 2.2 <i>Diagnostic tools</i> are operated following manufacturer's specification. 2.3 Faults and its causes are determined from diagnostic test results. 2.4 Diagnosis findings and recommendations, are reported according to industry criteria. 2.5 Safety practices are applied following Occupational Health and Safety (OSH) procedure.	2.1 Diagnostic symptoms and troubleshooting guide 2.2 Mensuration 2.3 Arithmetic operations 2.4 Use of measuring devices 2.5 Reporting procedures 2.6 Oil condition and viscosity 2.7 Noise 2.8 OSHS 2.9 Wearing of PPEs 2.10 Industry criteria 2.11 Different faults of continuously variable transmission 2.12 Diagnostic testing procedures for continuously variable transmission	2.1 Analyzing diagnostic symptoms 2.2 Interpreting information from manufacturer and workshop literature 2.3 Performing diagnostic tests 2.4 Inspecting continuously variable transmission 2.5 Comparing inspection results 2.6 Identifying faults and its causes 2.7 Reporting and recommendations findings 2.8 Applying safety practices 2.9 Mensuration skills 2.10 Applying arithmetic operations 2.11 Reporting diagnostic findings and make repair recommendations

3. Remove and replace continuously variable transmission assembly	3.1 Replacement information is sourced and interpreted following manufacturer's repair manual. 3.2 Replacement are carried out according to manufacturer specification workshop procedure and industry criteria. 3.3 Functionality test is carried out according to workplace procedures. 3.4 Documentation report is prepared based on analysis result.	3.1 Types, characteristics and operating principles of continuously variable transmission 3.2 Continuously variable transmission dismantling procedures 3.3 Continuously variable transmission component inspection, measuring and evaluation procedures 3.4 Component repair method 3.5 Mensuration 3.6 Arithmetic operation 3.7 Special service tools (SST) 3.8 Sourcing of continuously variable transmission components 3.9 OSHS 3.10 Wearing of PPEs 3.11 Replacement of continuously variable transmission components 3.12 3Rs 3.13 5S of Good Housekeeping 3.14 Tolerances and clearances of components of continuously variable transmission 3.15 Industry criteria 3.16 Assembly procedures 3.17 Post-assembly testing	3.1 Interpreting information from manufacturer and workshop literature 3.2 Measuring continuously variable transmission components 3.3 Using basic mathematical operations, including addition, subtraction, multiplication and division 3.4 Pull down skills 3.5 Arithmetic skills 3.6 Mensuration skills 3.7 Dismantling continuously variable transmission 3.8 Cleaning and evaluating continuously variable transmission components 3.9 Sourcing standard components 3.10 Using special service tools (SST) 3.11 Operating equipment 3.12 Replacing out of standard components 3.13 Assembling continuously variable transmission 3.14 Measuring tolerances and clearances of components of continuously variable transmission 3.15 Completing assembly of continuously
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ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
			variable transmission 3.16 Carrying out post-assembly testing 3.17 Applying safety practices
4. Complete work processes	4.1 Final inspection is conducted according to industry practices. 4.2 Vehicle is presented to client based on industry practices. 4.3 Work area is restored following 5S of good housekeeping. 4.4 Wastes are managed following environmental rules and regulations. 4.5 Tools and equipment are checked and stored according to workplace procedures. 4.6 Workplace documentation is processed according to workplace.	4.1 Final inspection procedure 4.2 Turn-over of vehicle 4.3 Accomplishment of repair order and other forms 4.3.1 Job done 4.4 OSHS 4.5 Wearing of PPEs 4.6 5S of Good Housekeeping 4.7 Waste management 4.8 Checking and storage of tools and equipment 4.9 Workplace documents	4.1 Filling-out workplace documentation. 4.2 Reporting diagnostic findings and make repair recommendations 4.3 Conducting final inspection 4.4 Performing vehicle turn-over 4.5 Restoring work area 4.6 Managing wastes 4.7 Checking and storing tools and equipment 4.8 Wearing of PPEs 4.9 Applying safety practices 4.10 Accomplishing workplace documents

RANGE OF VARIABLES

VARIABLE	RANGE
1. Job requirements	Job requirements may include: 1.1 Perform stall test 1.2 Perform Time lag test 1.3 Perform hydraulic pressure test 1.4 Initialization for continuously variable transmission 1.5 Replacement of continuously variable transmission 1.6 Diagnose available data trouble code 1.7 Replacement of power transmission control module (TCM)
2. Tools and equipment	Tools and equipment may include: 2.1 Tools: 2.1.1 Standard technician hand tools 2.1.2 Torque wrench 2.1.3 Wheel chocks 2.1.4 Multi-meter tester 2.1.5 Oil pressure gauge set with adapters 2.1.6 Diagnostic scan tools 2.2 Equipment: 2.2.1 Lifter 2.2.2 Air reel 2.2.3 Workshop compressor with air line 2.2.4 Electrical reel 2.2.5 Oil drain tank 2.2.6 Transmission jack 2.2.7 Diagnostic scan tool
3. Industry criteria	Industry criteria may include: 3.1 Manufacturer specifications 3.2 Repair manual 3.3 Workplace procedures 3.4 Safety and environmental requirements 3.5 Service history
4. Supplies	Supplies may include: 4.1 Rags 4.2 Continuously variable transmission fluid 4.3 Penetrating oil 4.4 Silicon sealant 4.5 Shock absorbing foam 4.6 PPEs 4.7 First aid kit
5. Diagnostic tests	Diagnostic tests may include: 5.1 Checking of oil condition and viscosity 5.2 Checking of voltage, continuity or resistance according to the standard 5.3 Checking of functionality and drivability 5.4 Checking of live data, freeze frame data and waveforms
6. Diagnostic tools	Diagnostic tools may include:

VARIABLE	RANGE
	6.1 Multi-meter tester 6.2 Oil pressure gauge set with adapters 6.3 Diagnostic scan tools
7. Functionality test	Functionality test may include: 7.1 Inspection of oil leak 7.2 Inspection of functionality and drivability 7.3 Road test 7.4 Perform stall test 7.5 Perform Time lag test 7.6 Perform hydraulic pressure test
8. Documentation report	Documentation report includes: 8.1 Accomplishment report 8.2 Recommendation report
9. Workplace documentation	Workplace documentation may include: 9.1 Repair order 9.2 Inspection form 9.3 Diagnostic form 9.4 Service history

EVIDENCE GUIDE

1. Critical aspects of competency	Assessment requires evidence that the candidate: 1.1 Prepared to diagnose continuously variable transmission. 1.1.1 Determined job requirements. 1.1.2 Selected and checked tools and equipment. 1.1.3 Applied safety practices. 1.2 Performed diagnosis. 1.2.1 Operated diagnostic tools. 1.2.2 Reported inspection findings and recommendations. 1.2.3 Applied safety practices. 1.3 Removed and replaced continuously variable transmission assembly. 1.3.1 Carried out repairs. 1.3.2 Carried out functionality test. 1.3.3 Applied safety practices. 1.4 Completed work processes. 1.4.1 Conducted final inspection. 1.4.2 Accomplished workplace documentation.
2. Resource implications	The following resources MUST be provided: 2.1 Actual or simulated workplace 2.2 Appropriate tools & equipment 2.3 Materials relevant to the activity 2.4 Manufacturer's repair manual 2.5 PPEs 2.6 Training vehicle 2.7 First aid kit
3. Method of assessment	Competency in this unit may be assessed through: 3.1 Demonstration/direct observation with oral questioning 3.2 Written exam 3.3 Portfolio
4. Context for assessment	4.1 Competency may be assessed individually in the actual workplace or simulation environment in TESDA accredited institutions

**UNIT OF COMPETENCY : DIAGNOSE AND REPAIR VEHICLE DYNAMICS
INTEGRATED MANAGEMENT**

UNIT CODE : ALT723365

UNIT DESCRIPTOR : This unit identifies the competence required to diagnose and repair faults in the hydraulic controlled brake systems of light vehicles. It involves preparing for the task, selecting the correct diagnostic procedure, carrying out the diagnosis and the repair, performing post-repair testing, and completing workplace processes and documentation.

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Prepare to diagnosis and repair activities	1.1 Safety practices are applied following OSHS. 1.2 Job requirements are determined from workplace instructions. 1.3 Diagnostic information is sourced and interpreted according to manufacturer specification and workplace procedures. 1.4 Hazards associated with the work are identified according to industry criteria. 1.5 Risks are managed based on industry criteria. 1.6 Tools and equipment are selected and checked for serviceability according to industry criteria . 1.7 Defective tools and equipment are reported following workplace procedures.	1.1 OSHS 1.2 Interpretation of job requirements 1.3 Sourcing out and interpretation of diagnostic information 1.4 Service/Repair manual 1.5 Functions and operation of hydraulic controlled brake system 1.6 Frictions 1.7 Work hazards 1.8 Troubleshooting guide 1.9 Tools, equipment and materials 1.10 Serviceability of tools and equipment 1.11 Parts and supplies 1.12 Mensuration 1.13 Arithmetic operation 1.14 Waste management	1.1 Applying safety practices 1.2 Interpreting job requirements from workplace instructions 1.3 Clarifying instructions 1.4 Locating appropriate sources of information 1.5 Identifying hazards 1.6 Managing risks 1.7 Selecting and checking tools and equipment 1.8 Reporting defective tools and equipment 1.9 Checking and reporting the availability of materials 1.10 Mensuration skills 1.11 Arithmetic skills

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
	1.8 <i>Parts and supplies</i> are sourced following workplace procedures.		

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
2. Perform diagnosis	2.1 <i>Diagnostic tools</i> are operated following manufacturer's specification. 2.2 <i>Diagnostic tests</i> are performed according to <i>industry criteria</i>. 2.3 Diagnostic options are analyzed and selected based manufacturer's manual and circumstances at hand. 2.4 Faults and its causes are determined based manufacturer's manual. 2.5 Diagnosis findings, and recommendations are reported according to industry criteria. 2.6 Safety practices are applied following Occupational Health and Safety Standard (OSHS).	2.1 Identification, function and operation of hydraulic controlled brake system 2.2 Troubleshooting guide 2.3 Types and application of brake fluids 2.4 Mensuration 2.5 Run out of rotor disc 2.6 Arithmetic operations 2.7 Use of measuring devices 2.8 Reporting procedures 2.9 Industry criteria 2.10 Inspection procedures for hydraulic controlled brake system, 2.11 Adjustment procedures: 2.11.1 adjusting brake pedal 2.11.2 bleeding brake fluid 2.11.3 parking brake adjustment 2.12 Procedure in accomplishing checklists 2.13 OSHS 2.14 Wearing of PPEs	2.1 Locating appropriate sources of information 2.2 Interpreting information from manufacturer and workshop literature 2.3 Measuring hydraulic controlled brake system components 2.4 Using basic mathematical operations, including addition and subtraction 2.5 Using measuring equipment 2.6 Interpreting measuring equipment scales 2.7 Performing diagnostic tests 2.8 Inspecting hydraulic controlled brake system 2.9 Comparing inspection results 2.10 Identifying faults and its causes 2.11 Applying safety practices 2.12 Mensuration skills 2.13 Applying arithmetic operations 2.14 Reporting diagnostic findings and make repair recommendations

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
3. Conduct repair	3.1 Repair information is sourced and interpreted following <i>industry criteria</i>. 3.2 Repair options and solutions are analyzed and selected based on manufacturer's manual and circumstances at hand. 3.3 Repairs are carried out according to <i>industry criteria</i>. 3.4 <i>Functionality test</i> is carried out to confirm fault rectification according to workplace procedures. 3.5 Rectification is done based on the results of functionality test. 3.6 <i>Documentation report</i> is prepared based on analysis result.	3.1 Identification, function and operation of hydraulic controlled brake system 3.2 Types and application of brake fluids 3.3 Repair and post-repair testing procedures for hydraulic controlled brake system 3.4 Tools, equipment, and materials 3.5 Mensuration 3.6 Service/Repair Manual 3.7 Different repair options 3.8 Arithmetic operations 3.9 Inspection procedures for hydraulic controlled brake system 3.10 Adjustment procedures 3.11 Procedure in accomplishing checklists 3.12 OSHS 3.13 Wearing of PPEs	3.1 Locating appropriate sources of information 3.2 Interpreting information from manufacturer and workshop literature 3.3 Measuring hydraulic controlled brake system components and using basic mathematical operations, including addition and subtraction 3.4 Interpreting measuring equipment scales 3.5 Using measuring equipment 3.6 Sourcing of information 3.7 Applying safety practices 3.8 Applying safety practices 3.9 Repairing hydraulic controlled brake system components 3.10 Performing post-repair testing 3.11 Accomplishing checklist

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
4. Complete work processes	4.1 Final inspection is conducted according to industry practices. 4.2 Vehicle is presented to client following industry protocol. 4.3 Work area is restored following 5S of good housekeeping. 4.4 Wastes are managed following environmental rules and regulations. 4.5 Tools and equipment are checked and stored according to workplace procedures. 4.6 Workplace documentation is processed according to workplace procedures.	4.1 Final inspection procedure 4.2 Turn-over of vehicle 4.3 Accomplishment of repair order and other forms 4.3.1 Job done 4.4 OSHS 4.5 Wearing of PPEs 4.6 3Rs 4.7 5S of Good Housekeeping 4.8 Waste management 4.9 Checking and storage of tools and equipment 4.10 Workplace documents	4.1 Conducting final inspection 4.2 Performing vehicle turn-over 4.3 Restoring work area 4.4 Managing wastes 4.5 Checking and storing tools and equipment 4.6 Wearing of PPEs 4.7 Applying safety practices 4.8 Accomplishing workplace documents

RANGE OF VARIABLES

VARIABLE	RANGE
1. Job requirements	Job requirements may include: 1.1 Diagnosis of ABS 1.2 Replacement of ABS components 1.3 Check performance of ABS
2. Tools and equipment	Tools and equipment may include: 2.1 Tools: 2.1.1 Standard technician hand tools 2.1.2 Torque wrench 2.1.3 Multi-meter tester 2.1.4 Diagnostic scan tools 2.2 Equipment: 2.2.1 Lifter 2.2.2 Air reel 2.2.3 Workshop compressor with air line 2.2.4 Electrical reel 2.2.5 Diagnostic scan tool
3. Industry criteria	Industry criteria may include: 3.1 Repair manual 3.2 Workplace procedures 3.3 Safety and environmental requirements 3.4 Service history
4. Parts and supplies	Parts and supplies may include: 4.1 Parts: 4.1.1 ABS wheel sensor 4.1.2 ABS actuator with module 4.2 Supplies: 4.2.1 Rags 4.2.2 Brake oil DOT3 or DOT4 4.2.3 Penetrating oil 4.2.4 Brake cleaner 4.2.5 PPEs 4.2.6 First aid kit
5. Diagnostic tests	Diagnostic tests may include: 5.1 Checking of braking distance 5.2 Checking of possible oil leaks 5.3 Checking ABS indicator 5.4 Checking of data list
6. Diagnostic tools	Diagnostic tools may include: 6.1 Multi-meter tester 6.2 Oil pressure gauge set with adapters 6.3 Diagnostic scan tools
7. Functionality test	Functionality test may include: 7.1 Checking of braking distance 7.2 Checking of possible oil leaks 7.3 Checking ABS indicator 7.4 Checking of data list
8. Documentation report	Documentation report Includes:

VARIABLE	RANGE
	8.1 Accomplishment 8.2 Recommendation
9. Workplace documentation	Workplace documentation may include: 9.1 Repair order 9.2 Inspection form 9.3 Diagnostic form 9.4 Service history

EVIDENCE GUIDE

1. Critical aspects of competency	Assessment requires evidence that the candidate: 1.1 Prepared to diagnose and repair activities. 1.1.1 Determined job requirements. 1.1.2 Selected and checked tools and equipment. 1.1.3 Applied safety practices. 1.2 Performed diagnosis. 1.2.1 Operated diagnostic tools. 1.2.2 Reported inspection findings and recommendations. 1.2.3 Applied safety practices. 1.3 Conducted repair activities. 1.3.1 Carried out repairs. 1.3.2 Carried out functionality test. 1.3.3 Applied safety practices. 1.4 Completed work processes. 1.4.1 Conducted final inspection. 1.4.2 Accomplished workplace documentation.
2. Resource implications	The following resources MUST be provided: 2.1 Actual or simulated workplace 2.2 Appropriate tools & equipment 2.3 Materials relevant to the activity 2.4 Manufacturer's repair manual 2.5 PPEs 2.6 Training vehicle 2.7 First aid kit
3. Method of assessment	Competency in this unit may be assessed through: 3.1 Demonstration/direct observation with oral questioning 3.2 Written exam 3.3 Portfolio
4. Context for assessment	4.1 Competency may be assessed individually in the actual workplace or simulation environment in TESDA accredited institutions

UNIT OF COMPETENCY : DIAGNOSE AND REPAIR ELECTRONIC 4-WHEEL DRIVE CONTROL SYSTEM

UNIT CODE : ALT723366

UNIT DESCRIPTOR : This unit identifies the competencies required to diagnose and repair of electronic 4-wheel drive control system. It involves identifying, evaluating, selecting, justifying, documenting and carrying out the most appropriate rectification method or variation. The unit includes the analysis of multisystem and intermittent faults which may be caused by operating in adverse conditions.

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Prepare to diagnose and repair electronic 4-wheel drive control system	1.1 Job requirements are determined from workplace instructions. 1.2 Diagnostic information is sourced and interpreted according to manufacturer specification and workplace procedures. 1.3 Hazards associated with the work are identified and risks are managed industry criteria. 1.4 Tools and equipment are selected and checked for serviceability according to industry criteria . 1.5 Defective tools and equipment are reported following workplace procedures. 1.6 Parts, supplies and materials are sourced following	1.1 OSHS 1.2 Waste management 1.3 Sourcing out and interpretation of repair information 1.4 Service/Repair manual 1.5 Tools, equipment and materials in inspecting and repairing drive lines and electronic 4-wheel control system 1.6 Interpretation of job requirements 1.7 Different job requirements 1.8 Serviceability of tools and equipment 1.9 Work hazards 1.10 Repair Order checklist 1.11 Identification and function of four wheel drive trains and electronic 4-wheel control system	1.1 Interpreting job requirements from workplace instructions 1.2 Clarifying instructions 1.3 Locating appropriate sources of information 1.4 Selecting and checking tools and equipment 1.5 Reporting defective tools and equipment 1.6 Checking and reporting the availability of materials 1.7 Applying safety practices

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
	workplace procedures. 1.7 Safety practices are applied following OSHS.		
2. Perform diagnosing	2.1 <i>Diagnostic tests</i> are performed according <i>industry criteria</i> . 2.2 Diagnostic tools are operated following manufacturer's specification. 2.3 Faults and their causes are determined according to industry criteria. 2.4 Diagnosis findings and recommendations are reported according to industry criteria. 2.5 Safety practices are applied following Occupational Health and Safety (OSH).	2.1 Inspection procedures for electronic 4-wheel drive control system 2.2 Identification and function of electronic 4-wheel drive control system 2.3 Mensuration 2.4 Arithmetic operations 2.5 Use of measuring devices 2.6 Reporting procedures 2.7 OSHS 2.8 Wearing of PPEs 2.9 Industry criteria	2.1 Interpreting information from manufacturer and workshop 2.2 Inspecting electronic 4-wheel drive control system 2.3 Comparing inspection results 2.4 Reporting and recommendations inspection findings 2.5 Applying safety practices 2.6 Mensuration skills 2.7 Applying arithmetic operations

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
3. Repair and implement performance improvement strategies	3.1 Repair information is sourced and interpreted following <i>industry criteria</i> . 3.2 Repair options and solutions are analyzed and selected based on manufacturer's manual circumstances at hand. 3.3 Repairs are carried out according to manufacturer specification workshop procedure and <i>industry criteria</i> . 3.4 <i>Functionality test</i> is carried out according to workplace procedures to confirm fault rectification. 3.5 Further problems detected as having been introduced during the repair process are rectified according to workplace procedures. 3.6 Documentation report is prepared based on analysis result.	3.1 Inspection procedures for electronic 4-wheel drive control system 3.2 Identification and function of electronic 4-wheel drive control system 3.3 Mensuration 3.4 Arithmetic operations 3.5 Use of measuring devices 3.6 Reporting procedures 3.7 OSHS 3.8 Wearing of PPEs 3.9 Industry criteria	3.1 Interpreting information from manufacturer and workshop 3.2 Inspecting electronic 4-wheel drive control system 3.3 Comparing inspection results 3.4 Reporting and recommendation s inspection findings 3.5 Applying safety practices 3.6 Mensuration skills 3.7 Applying arithmetic operations

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
4. Complete work processes	4.1 Final inspection is conducted following industry practice. 4.2 Vehicle is presented to client according to industry practice. 4.3 Work area is restored following 5S of good housekeeping. 4.4 Wastes are managed following environmental rules and regulations. 4.5 Tools and equipment are checked and stored according to workplace procedures. 4.6 Workplace documentation is processed according to workplace procedures.	4.1 Final inspection procedure: 4.1.1 Visual inspection 4.1.2 Checking of tightening of torque 4.2 Turn-over of vehicle 4.3 Accomplishment of repair order and other forms 4.3.1 Job done 4.4 OSHS 4.5 Wearing of PPEs 4.6 3Rs 4.7 5S of Good Housekeeping 4.8 Waste management 4.9 Checking and storage of tools and equipment 4.10 Workplace documents	4.1 Filling out workplace documentation 4.2 Conducting final inspection 4.3 Performing vehicle turn-over 4.4 Restoring work area 4.5 Managing wastes 4.6 Checking and storing tools and equipment 4.7 Wearing of PPEs 4.8 Applying safety practices

RANGE OF VARIABLES

VARIABLE	RANGE
1. Job requirements	Job requirements may include: 1.1 Inspection of front wheel axle 1.2 Removal/installation of front wheel axle 1.3 Inspection of rear wheel axle 1.4 Removal/installation of front wheel axle 1.5 Replacement of front and rear wheel hub bolts 1.6 Inspection of front drive shaft 1.7 Removal/installation front drive shaft 1.8 Disassembly/assembly of front drive shaft 1.9 Inspection of rear drive shaft 1.10 Removal/installation of rear drive shaft 1.11 Disassembly/assembly of rear drive shaft 1.12 Inspection of differential oil 1.13 Replacement of differential oil 1.14 Replacement of side gear oil seal 1.15 Removal/installation of rear differential 1.16 Removal/installation of rear differential mount rubber 1.17 Replacement of coupling components oil seal 1.18 Removal/installation of propeller shaft 1.19 Inspection of propeller shaft 1.20 Inspection of transfer case oil 1.21 Replacement of transfer case oil 1.22 Removal/installation of transfer case 1.23 Replacement of transfer case oil seal 1.24 Inspection of differential oil temperature sensor 1.25 Removal/installation of differential oil temperature sensor 1.26 Inspection of 4-wheel drive solenoid 1.27 Inspection of 4-wheel drive control module 1.28 Removal/installation of four-wheel drive control module 1.29 Removal/installation of coupling component 1.30 Configuration of 4-wheel drive control module 1.31 Calibration of coupling component 1.32 Replacement of powertrain control module 1.33 Diagnose available data trouble code 1.34 Inspection of 4-wheel drive switch 1.35 Inspection of 4-wheel drive actuator
2. Tools and equipment	Tools and equipment may include: 2.1 Tools: 2.1.1 Standard technician hand tools 2.1.2 Torque wrench 2.1.3 Dial gauge with magnetic base 2.1.4 Bench vise 2.1.5 Ball joint puller set (SST) 2.1.6 Bearing puller set (SST)

VARIABLE	RANGE
	2.1.7 Dust boot installer (SST) 2.1.8 Bearing installer (SST) 2.1.9 Support block (SST) 2.1.10 Boot clamp crimpers (SST) 2.1.11 Oil seal installer (SST) 2.1.12 Grease gun 2.1.13 Multi-meter tester 2.1.14 Diagnostic scan tools 2.1.15 Soldering iron 2.1.16 Snap ring pliers, 2 sets, (internal and external) 2.1.17 Sliding hammer (SST)*, 2 pcs 2.1.18 Tie rod end joint remover (SST)*, 2pcs *SST depends on the brand of the vehicle 2.2 Equipment: 2.2.1 Lifter 2.2.2 Air reel 2.2.3 Workshop compressor with air line 2.2.4 Electrical reel 2.2.5 Hydraulic press 2.2.6 Oil drain tank 2.2.7 Transmission jack 2.2.8 Diagnostic scan tool
3. Parts, supplies and materials	Parts, supplies and materials may include: 3.1 Supplies and materials 3.1.1 Rags 3.1.2 Gear oil 3.1.3 Penetrating oil 3.1.4 Grease 3.1.5 PPEs 3.1.6 Electrical tape 3.1.7 Soldering lead 3.1.8 Soldering paste 3.1.9 First aid kit 3.2 Parts: 3.2.1 4-wheel drive control module 3.2.2 4-wheel drive solenoid 3.2.3 4-wheel drive control module 3.2.4 4-wheel drive switch 3.2.5 4-wheel drive actuator
4. Diagnostic tests	Diagnostic tests may include: 4.1 Checking of oil and grease condition and viscosity 4.2 Checking of voltage, continuity or resistance according to the standard 4.3 Checking of functionality 4.4 Checking of possible oil and grease leaks 4.5 Checking and analyzing data list

VARIABLE	RANGE
5. Industry criteria	Industry criteria may include: 5.1 Repair manual 5.2 Workplace procedures 5.3 Safety and environmental requirements 5.4 Service history
6. Functionality test	Functionality test may include: 6.1 Inspection of oil and grease leaks 6.2 Inspection of functionality 6.3 Road test
7. Workplace documentation	Workplace documentation may include: 7.1 Repair order 7.2 Inspection form 7.3 Diagnostic form 7.4 Service history

EVIDENCE GUIDE

1. Critical aspects of competency	Assessment requires evidence that the candidate: 1.1 Prepare to diagnose and repair electronic 4-wheel drive control system. 1.1.1 Determined job requirements. 1.1.2 Sourced and interpreted diagnostic information 1.1.3 Identified hazards and managed. 1.1.4 Selected and checked tools and equipment for serviceability. 1.1.5 Reported defective tools and equipment. 1.1.6 Sourced parts, supplies and materials. 1.1.7 Applied safety practices. 1.2 Performed diagnosing. 1.2.1 Performed diagnostic tests. 1.2.2 Operated diagnostic tools. 1.2.3 Determined faults and their causes. 1.2.4 Reported diagnosis findings and recommendations. 1.2.5 Applied safety practices. 1.3 Repaired and implemented performance improvement strategies. 1.3.1 Sourced and interpreted repair information. 1.3.2 Analyzed and selected repair options and solutions. 1.3.3 Carried out repairs. 1.3.4 Carried out functionality test. 1.3.5 Rectified further problems detected during the repair process. 1.3.6 Prepared documentation report. 1.4 Completed work processes. 1.4.1 Conducted final inspection. 1.4.2 Presented vehicle to client. 1.4.3 Restored work area. 1.4.4 Managed wastes. 1.4.5 Checked and stored tools and equipment. 1.4.6 Processed workplace documentation.
2. Resource implications	The following resources MUST be provided: 2.1 Workplace: Real or simulated work area 2.2 Appropriate tools and equipment 2.3 Materials relevant to the activity 2.4 Manufacturer's repair manual 2.5 PPEs 2.6 Training vehicle 2.7 First aid kit
3. Method of assessment	Competency in this unit may be assessed through: 3.1 Demonstration/direct observation with Oral questioning 3.2 Written examination

4. Context for assessment	4.1 Competency may be assessed individually in the actual workplace or simulation environment in TESDA accredited institutions
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UNIT OF COMPETENCY : DIAGNOSE AND REPAIR 4-WHEEL DRIVE ASSEMBLY

UNIT CODE : ALT723367

UNIT DESCRIPTOR : This unit identifies the competence required to prepare to diagnose and repair 4-wheel drive assembly to completion.

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Prepare to diagnose and repair 4-wheel drive assembly	1.1 Safety practices are applied following OSHS. 1.2 Job requirements are determined from workplace instructions. 1.3 Diagnostic information is sourced and interpreted according to manufacturer specification and workplace procedures. 1.4 Hazards associated with the work are identified according to industry criteria. 1.5 Risks are managed based on industry criteria. 1.6 Tools and equipment are selected and checked for serviceability according to industry criteria. 1.7 Defective tools and equipment are reported following workplace procedures. 1.8 Parts, supplies and materials are sourced following	1.1 OSHS 1.2 Waste management 1.3 Sourcing out and interpretation of repair information 1.4 Service/Repair manual 1.5 Tools, equipment and materials in inspecting and repairing 4-wheel drive assembly 1.6 Interpretation of job requirements 1.7 Different job requirements 1.8 Serviceability of tools and equipment 1.9 Work hazards 1.10 Repair Order checklist 1.11 Identification and function of 4-wheel drive assembly	1.1 Interpreting job requirements from workplace instructions 1.2 Clarifying instructions 1.3 Locating appropriate sources of information 1.4 Selecting and checking tools and equipment 1.5 Reporting defective tools and equipment 1.6 Checking and reporting the availability of materials 1.7 Applying safety practices

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
	workplace procedures.		
2. Diagnose 4-wheel drive assembly	2.1 Diagnostic tests are performed according to <i>industry criteria</i>. 2.2 Diagnostic tools are operated following manufacturer's specification. 2.3 Faults and their causes are determined based on manuals. 2.4 Diagnosis findings, and recommendations are reported according to industry criteria. 2.5 Safety practices are applied following Occupational Health and Safety (OSH).	2.1 Inspection procedures for electronic 4-wheel drive assembly 2.2 Identification and function of electronic 4-wheel drive control system 2.3 Mensuration 2.4 Arithmetic operations 2.5 Use of measuring devices 2.6 Reporting procedures 2.7 OSHS 2.8 Wearing of PPEs 2.9 Industry criteria	2.1 Interpreting information from manufacturer and workshop 2.2 Inspecting electronic 4-wheel drive control system 2.3 Comparing inspection results 2.4 Reporting and recommendation s inspection findings 2.5 Applying safety practices 2.6 Mensuration skills 2.7 Applying arithmetic operations

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
3. Repair 4-wheel drive assembly	3.1 Repair information is sourced and interpreted following industry criteria . 3.2 Repair options and solutions are analyzed and selected based on manufacturer's manual circumstances at hand. 3.3 Repairs are carried out according to manufacturer specification workshop procedure and industry criteria . 3.4 Functionality test is carried out according to workplace procedures to confirm fault rectification according to manual. 3.5 Further problems detected as having been introduced during the repair process are rectified following manuals. 3.6 Documentation report is prepared based on analysis result.	3.1 Inspection procedures for electronic 4-wheel drive control system 3.2 Identification and function of 4-wheel drive assembly 3.3 Mensuration 3.4 Arithmetic operations 3.5 Use of measuring devices 3.6 Reporting procedures 3.7 OSHS 3.8 Wearing of PPEs 3.9 Industry criteria	3.1 Interpreting repair information from manufacturer and workshop 3.2 Analyzing and selecting repair options and solutions for 4-wheel drive assembly 3.3 Carrying out repair of 4-wheel drive assembly 3.4 Performing functionality test 3.5 Confirming fault rectification 3.6 Rectifying detected problems 3.7 Reporting and recommendations inspection findings 3.8 Applying safety practices 3.9 Mensuration skills 3.10 Applying arithmetic operations

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
4. Perform activities to completion	4.1 Final inspection is conducted based on industry practice. 4.2 Vehicle is presented to client following industry procedure. 4.3 Work area is restored following 5S of Good Housekeeping. 4.4 Wastes are managed following environmental rules and regulations. 4.5 Tools and equipment are checked and stored according to workplace procedures. 4.6 Workplace documentation is processed according to workplace procedures.	4.1 Final inspection procedure: 4.1.1 Visual inspection 4.1.2 Checking of tightening of torque 4.2 Turn-over of vehicle 4.3 Accomplishment of repair order and other forms 4.3.1 Job done 4.4 OSHS 4.5 Wearing of PPEs 4.6 3Rs 4.7 5S of Good Housekeeping 4.8 Waste management 4.9 Checking and storage of tools and equipment 4.10 Workplace documents	4.1 Filling out workplace documentation 4.2 Conducting final inspection 4.3 Performing vehicle turn-over 4.4 Restoring work area 4.5 Managing wastes 4.6 Checking and storing tools and equipment 4.7 Wearing of PPEs 4.8 Applying safety practices

RANGE OF VARIABLES

VARIABLE	RANGE
1. Job requirements	Job requirements may include: 1.1 Inspection of front wheel axle 1.2 Removal/installation of front wheel axle 1.3 Inspection of rear wheel axle 1.4 Removal/installation of front wheel axle 1.5 Replacement of front and rear wheel hub bolts 1.6 Inspection of front drive shaft 1.7 Removal/installation front drive shaft 1.8 Disassembly/assembly of front drive shaft 1.9 Inspection of rear drive shaft 1.10 Removal/installation of rear drive shaft 1.11 Disassembly/assembly of rear drive shaft 1.12 Inspection of differential oil 1.13 Replacement of differential oil 1.14 Replacement of side gear oil seal 1.15 Removal/installation of rear differential 1.16 Removal/installation of rear differential mount rubber 1.17 Replacement of coupling components oil seal 1.18 Removal/installation of propeller shaft 1.19 Inspection of propeller shaft 1.20 Replacement of transfer case oil 1.21 Removal/installation of transfer case 1.22 Replacement of transfer case oil seal 1.23 Removal/installation of coupling component 1.24 Calibration of coupling component
2. Tools and equipment	Tools and equipment may include: 2.1 Tools: 2.1.1 Standard technician hand tools 2.1.2 Torque wrench 2.1.3 Dial gauge with magnetic base 2.1.4 Bench vise 2.1.5 Ball joint puller set (SST) 2.1.6 Bearing puller set (SST) 2.1.7 Dust boot installer (SST) 2.1.8 Bearing installer (SST) 2.1.9 Support block (SST) 2.1.10 Boot clamp crimpers (SST) 2.1.11 Oil seal installer (SST) 2.1.12 Grease gun 2.1.12.1 Multimeter tester 2.1.12.2 Diagnostic scan tools 2.1.12.3 Soldering iron 2.2 Equipment: 2.2.1 Lifter 2.2.2 Air reel

VARIABLE	RANGE
	2.2.3 Workshop compressor with air line 2.2.4 Electrical reel 2.2.5 Hydraulic press 2.2.6 Oil drain tank 2.2.7 Transmission jack 2.2.8 Diagnostic scan tool
3. Parts, supplies and materials	Parts, supplies and materials may include: 3.1 Supplies and materials: 3.1.1 Rags 3.1.2 Gear oil 3.1.3 Penetrating oil 3.1.4 Grease 3.1.5 PPEs 3.1.6 Vehicle protective kit: 3.1.6.1 Seat cover 3.1.6.2 Steering wheel cover 3.1.6.3 Fender cover 3.1.6.4 Shifting knob / lever cover 3.1.6.5 Paper matting 3.1.7 First aid kit 3.2 Parts: 3.2.1 Front wheel axle 3.2.2 Rear wheel axle 3.2.3 Front and rear wheel hub bolts 3.2.4 Front drive shaft 3.2.5 Rear drive shaft 3.2.6 Transfer case 3.2.7 Differential case
4. Diagnostic tests	Diagnostic tests may include: 4.1 Checking of oil and grease condition and viscosity 4.2 Checking of voltage, continuity or resistance according to the standard 4.3 Checking of functionality 4.4 Checking of possible oil and grease leaks
5. Industry criteria	Industry criteria may include: 5.1 Repair manual 5.2 Workplace procedures 5.3 Safety and environmental requirements 5.4 Service history
6. Repair of components of 4-wheel drive assembly	Repair of components of 4-wheel drive assembly may include: 6.1 Repair mechanical components of transfer case 6.2 Repair of electronic control system 6.3 Overhaul of transfer case
7. Functionality test	Functionality test may include: 7.1 Road test 7.2 Inspection of vibration and noise 7.3 Inspection of oil leak

VARIABLE	RANGE
8. Workplace documentation	Workplace documentation may include: 8.1 Repair order 8.2 Inspection form 8.3 Diagnostic form 8.4 Service history

EVIDENCE GUIDE

1. Critical aspects of competency	Assessment requires evidence that the candidate: 1.1 Prepared to diagnose and repair 4-wheel drive assembly. 1.1.1 Applied safety practices. 1.1.2 Determined job requirements. 1.1.3 Sourced and interpreted diagnostic information. 1.1.4 Identified hazards associated with the work. 1.1.5 Managed risks. 1.1.6 Selected and checked tools and equipment. 1.1.7 Reported defective tools and equipment. 1.1.8 Sourced parts, supplies and materials. 1.2 Diagnosed 4-wheel drive assembly. 1.2.1 Performed diagnostic tests. 1.2.2 Operated diagnostic tools. 1.2.3 Determined faults and their causes. 1.2.4 Reported diagnosis findings and recommendations. 1.2.5 Applied safety practices. 1.3 Repaired 4-wheel drive assembly. 1.3.1 Sourced and interpreted repair information. 1.3.2 Analyzed and selected repair options and solutions. 1.3.3 Carried out repairs. 1.3.4 Carried out functionality test. 1.3.5 Rectified further problems detected during the repair process. 1.3.6 Prepared documentation report. 1.4 Performed activities to completion. 1.4.1 Conducted final inspection. 1.4.2 Presented vehicle to client. 1.4.2 Restored work area. 1.4.3 Managed wastes. 1.4.4 Checked and stored tools and equipment. 1.4.5 Processed workplace documentation.
2. Resource implications	The following resources MUST be provided: 2.1 Workplace: Real or simulated work area 2.2 Appropriate tools & equipment 2.3 Materials relevant to the activity 2.4 Manufacturer's repair manual 2.5 PPEs 2.6 Training vehicle 2.7 First aid kit
3. Method of assessment	Competency in this unit must be assessed through: 3.1 Demonstration/direct observation with Oral questioning 3.2 Written exam
4. Context for assessment	4.1 Competency may be assessed individually in the actual workplace or simulation environment in TESDA accredited institutions

UNIT OF COMPETENCY : DIAGNOSE AND REPAIR ELECTRIC POWER STEERING

UNIT CODE : ALT723368

UNIT DESCRIPTOR : This unit identifies the competence required to diagnose complex faults in electric power steering and determine the repair action necessary to restore system performance. It involves confirming the existence of a fault, choosing the diagnostic procedure and tools, applying the diagnostic procedure, reporting conclusions and making repair recommendations.

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
1. Prepare for diagnose and repair of electric power steering	1.1 Job requirements are determined from workplace instructions. 1.2 Diagnostic information is sourced and interpreted according to manufacturer specification and workplace procedures. 1.3 Hazards associated with the work are identified and risks are managed industry criteria. 1.4 Tools and equipment are selected and checked for serviceability according to industry criteria. 1.5 Defective tools and equipment are reported following workplace procedures. 1.6 Parts, supplies and materials are sourced following	1.1 OSHS 1.2 Waste management 1.3 Tools, equipment and materials 1.4 Troubleshooting guide 1.5 Mensuration 1.6 Arithmetic operation 1.7 Functions and operation of electric power steering 1.8 Repair manual 1.9 Work hazards 1.10 Sourcing out and interpretation of diagnostic information 1.11 Service/Repair manual 1.12 Interpretation of job requirements 1.13 Different job requirements 1.14 Serviceability of tools and equipment	1.1 Interpreting job requirements from workplace instructions 1.2 Clarifying instructions 1.3 Locating appropriate sources of information 1.4 Selecting and checking tools and equipment 1.5 Mensuration skills 1.6 Arithmetic skills 1.7 Reporting non-serviceable tools and equipment 1.8 Checking and reporting the availability of materials 1.9 Applying safety practices

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
	workplace procedures. 1.7 Safety practices are applied following OSHS.		
2. Diagnose electric power steering	2.1 Diagnostic method sequence, tests and testing processes are selected from the range of available options. 2.2 Tools, supplies and materials required to support the diagnostic process are selected for use. 2.3 <i>Diagnostic tests</i> are performed according to <i>industry criteria</i> . 2.4 Diagnostic scan tools are prepared and operated following manufacturer's specification. 2.5 Faults and their causes are determined based on diagnostic results. 2.6 Diagnosis findings, and recommendations are reported according to industry criteria. 2.7 Safety practices are applied following Occupational Health and Safety (OSH).	2.1 Identification, function and operation of electric power steering 2.2 Troubleshooting table 2.3 Mensuration 2.4 Arithmetic operations 2.5 Use of measuring devices 2.6 Reporting procedures 2.7 Industry criteria 2.8 Inspection procedures for electric power steering 2.9 Procedure in accomplishing checklists 2.10 OSHS 2.11 Wearing of PPEs	2.1 Locating appropriate sources of information. 2.2 Interpreting information from manufacturer and workshop literature. 2.3 Measuring steering components and 2.4 Using basic mathematical operations, including addition and subtraction 2.5 Interpreting measuring equipment scales. 2.6 Using measuring equipment, such as hydraulic pressure gauge and torque wrench 2.7 Performing diagnostic tests 2.8 Inspecting steering system 2.9 Comparing inspection results 2.10 Identifying faults and its causes 2.11 Applying safety practices 2.12 Reporting diagnostic findings and make repair recommendations

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
3. Repair electric power steering	3.1 Conclusions are drawn from findings and provided to appropriate personnel and customer to confirm further action to be taken. 3.2 Repair information is sourced and interpreted following industry criteria. 3.3 Repair options and solutions are analyzed and selected based on manufacturer's manual circumstances at hand. 3.4 Repairs are carried out according to manufacturer specification workshop procedure and industry criteria. 3.5 Functionality test is carried out according to workplace procedures. 3.6 Any problems detected as having been introduced during the repair process are rectified following the manufacturer's manuals. 3.7 Documentation report is prepared based on analysis result.	3.1 Identification, function and operation of electric power steering 3.2 Repair and post-repair testing procedures for electric power steering 3.3 Tools, equipment and materials 3.4 Mensuration 3.5 Service/Repair Manual 3.6 Different repair options 3.7 Arithmetic operations 3.8 Inspection procedures for electric power steering 3.9 Adjustment procedures 3.10 Procedure in accomplishing checklists 3.11 OSHS 3.12 Wearing of PPEs	3.1 Monitoring service and adjustments 3.2 Locating appropriate sources of information efficiently. 3.3 Interpreting information from manufacturer and workshop literature. 3.4 Measuring electric power steering components 3.5 Using basic mathematical operations, including addition and subtraction 3.6 Interpreting measuring equipment scales. 3.7 Using measuring equipment 3.8 Sourcing of information 3.9 Applying safety practices 3.10 Repairing electric power steering 3.11 Performing post-repair testing

ELEMENT	PERFORMANCE CRITERIA <i>Italicized terms</i> are elaborated in the Range of Variables	REQUIRED KNOWLEDGE	REQUIRED SKILLS
4. Complete work processes	4.1 Vehicle is presented ready to be repaired and returned to the customer. 4.2 Work area is cleaned, waste and non-recyclable materials are disposed of, and recyclable material is collected. 4.3 Tools and equipment are checked and stored according to workplace procedures.	4.1 Final inspection procedure 4.2 Turn-over of vehicle 4.3 Accomplishment of repair order and other forms 4.3.1 Job done 4.4 OSHS 4.5 Wearing of PPEs 4.6 3Rs 4.7 5S of Good Housekeeping 4.8 Waste management 4.9 Checking and storage of tools and equipment 4.10 Workplace documents	4.1 Reporting inspection findings and make repair recommendations. 4.2 Conducting final inspection 4.3 Performing vehicle turn-over 4.4 Restoring work area 4.5 Managing wastes 4.6 Checking and storing tools and equipment 4.7 Wearing of PPEs 4.8 Applying safety practices 4.9 Accomplishing workplace documents

RANGE OF VARIABLES

VARIABLE	RANGE
1. Job requirements	Job requirements may include: 1.1 Inspect electric power steering control module 1.2 Inspect electric power steering control motor 1.3 Analyze data list 1.4 Check malfunction indicator light (EPS) 1.5 Diagnose electric power steering system (EPS) 1.6 Replace EPS control module 1.7 Replace EPS control motor 1.8 Perform initialization
2. Tools and equipment	Tools and equipment may include: 2.1 Tools: 2.1.1 Multi-meter tester 2.2 Equipment: 2.2.1 Lifter 2.2.2 Diagnostic scan tools
3. Industry criteria	Industry criteria may include: 3.1 Repair manual 3.2 Workplace procedures 3.3 Safety and environmental requirements 3.4 Service history
4. Materials	Materials may include: 4.1 Rags 4.2 PPE's 4.3 Penetrating oil 4.4 Electronic grease 4.5 First aid kit
5. Diagnostic tests	Diagnostic tests may include: 5.1 Checking of voltage, continuity or resistance according to the standard 5.2 Checking of functionality 5.3 Physical damage 5.4 Noise and vibration 5.5 Alignment of steering wheel
6. Functionality test	Functionality test includes: 6.1 Inspect steering free play 6.2 Inspect the steering resistance 6.3 Check steering system for noise
7. Repairs	Repairs may include: 7.1 Erasing diagnostic trouble code (DTC) 7.2 Remove and replace motor 7.3 Remove and replace harness 7.4 Remove and replace module
8. Documentation report	Documentation report includes: 8.1 Accomplishment 8.2 Recommendation

EVIDENCE GUIDE

1. Critical aspects of competency	Assessment requires evidence that the candidate: 1.1 Prepared for diagnosis and repair of electric power steering. 1.1.1 Determined job requirements. 1.1.2 Sourced and interpreted diagnostic information. 1.1.3 Identified hazards and managed risks. 1.1.4 Selected and checked tools and equipment. 1.1.5 Reported defective tools and equipment. 1.1.6 Sourced parts, supplies and materials. 1.1.7 Applied safety practices. 1.2 Diagnosed electric power steering. 1.2.1 Selected diagnostic method sequence, tests and testing processes. 1.2.2 Selected tools, supplies and materials required to diagnostic process. 1.2.3 Performed diagnostic tests. 1.2.4 Prepared and operated diagnostic scan tools. 1.2.5 Determined faults and their causes. 1.2.6 Reported diagnosis findings and recommendations. 1.2.7 Applied safety practices. 1.3 Repaired electric power steering. 1.3.1 Drawn conclusions and provided to appropriate personnel and customer. 1.3.2 Sourced and interpreted repair information. 1.3.3 Analyzed and selected repair options and solutions. 1.3.4 Carried out repairs. 1.3.5 Carried out functionality test. 1.3.6 Rectified any problems detected during the repair process. 1.3.7 Prepared documentation report. 1.4 Completed work processes. 1.4.1 Presented vehicle is ready to be repaired and returned to customer. 1.4.2 Cleaned work area, disposed of waste and non-recyclable materials, and collected recyclable material. 1.4.3 Checked and stored tools and equipment.
2. Resource implications	The following resources MUST be provided: 2.1 Workplace: Real or simulated work area 2.2 Appropriate Tools & equipment 2.3 Materials relevant to the activity 2.4 Manufacturer's repair manual 2.5 PPEs 2.6 Training vehicle 2.7 First aid kit

3. Method of assessment	Competency in this unit may be assessed through: 3.1 Demonstration/direct observation with oral questioning 3.2 Written examination 3.3 Portfolio
4. Context for assessment	4.1 Competency may be assessed individually in the actual workplace or simulation environment in TESDA accredited institutions

SECTION 3 TRAINING ARRANGEMENTS

These guidelines are set to provide the Technical and Vocational Education and Training (TVET) providers with information and other important requirements to consider when designing training programs for **AUTOMOTIVE DIAGNOSIS (CHASSIS) NC III**.

3.1 CURRICULUM DESIGN

TESDA shall provide the training on the development of competency-based curricula to enable training providers develop their own curricula with the components mentioned below.

Delivery of knowledge requirements for the basic, common and core units of competency specifically in the areas of mathematics, science/technology, communication/language and other academic subjects shall be contextualized. To this end, TVET providers shall develop a Contextual Learning Matrix (CLM) to accompany the curricula.

Course Title: **AUTOMOTIVE DIAGNOSIS (CHASSIS)** NC Level **NC III**

Nominal Training Duration:

40	Hours (Basic Competencies)
162	Hours (Common Competencies)
170	Hours (Core Competencies)
372	TOTAL HOURS
150	SIL

Course Description:

This course is designed to enhance the knowledge, skills and attitudes of an individual in the field of automotive diagnosis in accordance with industry standards. It covers core competencies such as: diagnose and repair of electronically controlled transmission; diagnose and repair of continuously variable transmission; diagnose and repair of vehicle dynamics integrated management; diagnose and repair of electronic 4-wheel drive control system; diagnose and repair of electronic 4-wheel drive assembly; and diagnose and repair electric power steering.

This course is also designed to enhance the basic and common knowledge, skills and attitudes of an individual in the field of automotive diagnosis.

To obtain this, all units prescribed for this qualification must be achieved.

BASIC COMPETENCIES
40 Hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
1. Lead workplace communication	1.1 Communicate information about workplace processes	1.1.1 Read ○ Effective verbal communication methods ○ Sources of information	● Lecture	● Written Test	2 Hours
		1.1.2 Practice organizing information	● Demonstration	● Observation	
		1.1.3 Identify organization requirements for written and electronic communication methods	● Lecture	● Written Test	
		1.1.4 Follow organization requirements for the use of written and electronic communication methods	● Demonstration ● Practical exercises	● Observation	
		1.1.5 Perform exercises on understanding and conveying intended meaning scenario	● Demonstration ● Role Play	● Observation	
	1.2 Lead workplace discussions	1.2.1 Describe: ● Organizational policy on production, quality and safety ● Goals/ objectives and action plan setting	● Group discussion	● Oral evaluation	2 Hours
		1.2.2 Read ● Effective verbal communication methods	● Lecture	● Written Test	
		1.2.3 Prepare/set action plans based on organizational goals and objectives	● Demonstration	● Observation	

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
	1.3 Identify and communicate issues arising in the workplace	1.2.4 Describe: Organizational policy in dealing with issues and problems	● Group discussion	● Oral evaluation	2 Hours
		1.2.5 Read Effective verbal communication methods	● Lecture	● Written Test	
2. Lead small teams	2.1 Provide team leadership	2.1.1 Discussion of Company policies and procedures 2.1.2 Read web pages on situational leadership 2.1.3 Role play on situational leadership	● Group work ● Role Play ● Lecture/ Discussion ● Individual Work	● Role Play ● Written Test	1 hour
	2.2 Assign responsibilities	2.2.1 Read web pages on performance management 2.2.2 Case study on allocating roles and responsibilities based on competencies of current staff	● Individual Work ● Case Study	● Role Play ● Written Test	1 hour
	2.3 Set performance expectations for team members	2.3.1 Role play to communicate performance expectations with staff 2.3.2 Discussion on performance issues	● Lecture/ Discussion ● Role Play	● Role Play ● Written Test	1 hour
	2.4 Supervise team performance	2.4.1 Discussion on performance monitoring 2.4.2 Role play on providing feedback on performance 2.4.3 Role play on performance coaching 2.4.4 Discussion on keeping the team informed of team performance	● Lecture/ Discussion ● Role Play ● Case Study	● Role Play ● Written Test	1 hour

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		2.4.5 Case study on Team performance monitoring and feedback			
3. Apply critical thinking and problem-solving techniques in the workplace	3.1 Examine specific workplace strategies	3.1.1 Show thorough knowledge and understanding of the process, normal operating parameters, and product quality to recognize non-standard situations 3.1.2 Show mastery of the current industry hardware and software products and services 3.1.3 Discuss process of identification of fundamental causes of specific workplace challenges 3.1.4 Show mastery of knowledge and understanding of the process, normal operating parameters, and product quality to recognize non-standard situations • Relevant equipment and operational processes • Enterprise goals, targets and measures • Enterprise quality OHS and environmental requirement • Enterprise information systems and data collation • Industry codes and standards	• Group discussion • Lecture • Demonstration • Role playing	• Case Formulation • Life Narrative Inquiry (Interview) • Standardized test	1 hour

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
	3.2 Analyze the causes of specific workplace challenges	3.2.1 Show thorough knowledge and understanding of the process, normal operating parameters, and product quality to recognize non-standard situations 3.2.2 Show mastery of the current industry hardware and software products and services 3.2.3 Discuss process of identification of fundamental causes of specific workplace challenges 3.2.4 Show mastery of knowledge and understanding of the process, normal operating parameters, and product quality to recognize non-standard situations • Relevant equipment and operational processes • Enterprise goals, targets and measures • Enterprise quality OHS and environmental requirement • Enterprise information systems and data collation • Industry codes and standards 3.2.5 Identify extent and causes of specific challenges in the workplace 3.2.6 Use of range of analytical problem-solving techniques	• Group discussion • Lecture • Demonstration • Role playing	• Case Formulation • Life Narrative Inquiry (Interview) • Standardized test	1 hour

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		3.2.7 Formulate clear-cut findings on the nature of each identified workplace challenges			
	3.3 Formulate resolutions to specific workplace challenges	3.3.1 Show thorough knowledge and understanding of the process, normal operating parameters, and product quality to recognize non-standard situations 3.3.2 Show mastery of the current industry hardware and software products and services 3.3.3 Discuss process of identification of fundamental causes of specific workplace challenges 3.3.4. Show mastery of knowledge and understanding of the process, normal operating parameters, and product quality to recognize non-standard situations • Relevant equipment and operational processes • Enterprise goals, targets and measures • Enterprise quality OHS and environmental requirement • Enterprise information systems and data collation • Industry codes and standards 3.3.5 Identify extent and causes of specific challenges in the workplace 3.3.6 Use of range of analytical problem-solving techniques	• Group discussion • Lecture • Demonstration • Role playing	• Case Formulation • Life Narrative Inquiry (Interview) • Standardized test	1 hour

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		3.3.7 Formulate clear-cut findings on the nature of each identified workplace challenges 3.3.8 Discuss strategies on devising, communicating, implementing and evaluating strategies and techniques in addressing specific workplace challenges			
	3.4 Implement action plans and communicate results	3.4.1 Identify extent and causes of specific challenges in the workplace 3.4.2 Use of range of analytical problem-solving techniques 3.4.3 Formulate clear-cut findings on the nature of each identified workplace challenges 3.4.4 Discuss strategies on devising, communicating, implementing and evaluating strategies and techniques in addressing specific workplace challenges	• Group discussion • Lecture • Demonstration • Role playing	• Case Formulation • Life Narrative Inquiry (Interview) • Standardized test	1 hour
4. Work in a Diverse Environment	4.1 Develop an individual's cultural awareness and sensitivity	4.1.1 Show understanding of cultural diversity in the workplace 4.1.2 Recognize norms of behavior for interacting and dialogue with specific groups (e. g., Muslims and other non-Christians, non-Catholics, tribes/ethnic groups, foreigners) 4.1.3 Demonstrate different methods of verbal and non-verbal communication in a multicultural setting	• Small Group Discussion • Interactive Lecture • Brainstorming • Demonstration • Role-playing	• Demonstration or simulation with oral questioning • Group discussions and interactive activities • Case studies/problems involving workplace diversity issues	1 hour

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		4.1.4 Apply cross-cultural communication skills (i.e. different business customs, beliefs, communication strategies) 4.1.5 Show affective skills – establishing rapport and empathy, understanding, etc. 4.1.6 Demonstrate openness and flexibility in communication 4.1.7 Recognize diverse groups in the workplace and community as defined by divergent culture, religion, traditions and practices		• Written examination • Role Playing	
	4.2 Work effectively in an environment that acknowledges and values cultural diversity	4.2.1 Explain the value of diversity in the economy and society in terms of 4.2.2 Workforce development 4.2.3 Discuss the importance of inclusiveness in a diverse environment 4.2.4 Discuss the importance of shared vision and understanding of and commitment to team, departmental, and organizational goals and objectives 4.2.5 Identify and exhibit strategies for customer service excellence 4.2.6 Demonstrate cross-cultural communication skills and active listening 4.2.7 Recognize diverse groups in the workplace and community as	• Small Group Discussion • Interactive Lecture • Brainstorming • Demonstration • Role-playing	• Demonstration or simulation with oral questioning • Group discussions and interactive activities • Case studies/problems involving workplace diversity issues • Written examination • Role Playing	1 hour

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		defined by divergent culture, religion, traditions and practices 4.2.8 Demonstrate collaboration skills			
	4.3 Identify common issues in a multicultural and diverse environment	4.3.1 Explain the value, and leverage of cultural diversity 4.3.2 Discuss the inclusivity and conflict resolution 4.3.3 Describe the workplace harassment 4.3.4 Explain the change management and cite ways to overcome resistance to change 4.3.5 Demonstrate advanced strategies for customer service excellence 4.3.6 Address diversity-related conflicts in the workplace 4.3.7 Eliminate discriminatory behavior towards customers and co-workers 4.3.8 Utilize change management policies in the workplace	• Small Group Discussion • Interactive Lecture • Brainstorming • Demonstration • Role-playing	• Demonstration or simulation with oral questioning • Group discussions and interactive activities • Case studies/problems involving workplace diversity issues • Written examination • Role Playing	1 hour
5. Propose methods of applying learning and innovation in the organization	5.1 Assess work procedures, processes and systems in terms of innovative practices	5.1.1 Show mastery of the following practical concepts (e.g., 7 habits of highly effective people, character strengths that foster learning and innovation, five minds of the future, adaptation concepts and transtheoretical model of behavior change) 5.1.2 Demonstrate collaboration and networking skills 5.1.3 Show basic skills in research	• Interactive Lecture • Appreciative Inquiry • Demonstration • Group work	• Psychological and behavioral Interviews • Performance Evaluation • Life Narrative Inquiry • Review of portfolios of evidence and third-party	1 hour

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		5.1.4 Generate practical insights on how to improve organizational procedures, processes and systems		workplace reports of on-the-job performance. ● Standardized assessment of character strengths and virtues applied	
	5.2 Generate practical action plans for improving work procedures, processes	5.2.1 Show mastery of the following practical concepts (e.g., 7 habits of highly effective people, character strengths that foster learning and innovation, five minds of the future, adaptation concepts and transtheoretical model of behavior change) 5.2.2 Demonstrate collaboration and networking skills 5.2.3 Show basic skills in research 5.2.4 Generate practical insights on how to improve organizational procedures, processes and systems 5.2.5 Set up action plans on how to apply innovative procedures in the organization 5.2.6 Set up action plans on how to apply innovative procedures in the organization 5.2.7 Generate practical insights on how to improve organizational	● Interactive Lecture ● Appreciative Inquiry ● Demonstration ● Group work	● Psychological and behavioral Interviews ● Performance Evaluation ● Life Narrative Inquiry ● Review of portfolios of evidence and third-party workplace reports of on-the-job performance. ● Standardized assessment of character strengths and virtues applied	1 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		procedures, processes and systems			
	5.3 Evaluate the effectiveness of the proposed action plans	5.3.1 Show mastery of the following practical concepts (e.g., 7 habits of highly effective people, character strengths that foster learning and innovation, five minds of the future, adaptation concepts and transtheoretical model of behavior change) 5.3.2 Demonstrate collaboration and networking skills 5.3.3 Show basic skills in research 5.3.4 Generate practical insights on continuous improvement	• Interactive Lecture • Appreciative Inquiry • Demonstration • Group work	• Psychological and behavioral Interviews • Performance Evaluation • Life Narrative Inquiry • Review of portfolios of evidence and third-party workplace reports of on-the-job performance. • Standardized assessment of character strengths and virtues applied	1 hour
6. Use information systematically	6.1 Use technical information systems	6.1.1 Lecture and discussion on: • Application in collating information • Procedures for inputting, maintaining and archiving information • Guidance to people who need to find and use information 6.1.2 Organizing information into a suitable form for reference and use	• Lecture • Group Discussion • Hands on • Demonstration	• Oral evaluation • Written Test • Observation • Presentation	4 Hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		6.1.3 Classify stored information for identification and retrieval 6.1.4 Operate the technical information system by using agreed procedures			
	6.2 Apply information technology (IT) systems	6.2.1 Lecture and discussion on: • Attributes and limitations of available software tool • Procedures and work instructions for the use of IT • Operational requirements for IT systems • Sources and flow paths of data • Security systems and measures that can be used • Methods of entering and processing information 6.2.2. Use procedures and work instructions for the use of IT 6.2.3 Extract data and format reports 6.2.4 Use WWW applications	• Lecture • Group Discussion • Self-paced handout/ module • Hands on • Demonstration	• Oral evaluation • Written Test • Observation • Presentation	2 Hours
	6.3 Edit, format and check information	6.3.1 Lecture and discussion on: • Basic file-handling techniques • Techniques in checking documents • Techniques in editing and formatting • Proof reading techniques 6.3.2 Use different techniques in checking documents 6.3.3 Edit and format information applying different techniques	• Lecture • Group Discussion • Self-paced handout/ module • Hands on • Demonstration	• Oral evaluation • Written Test • Observation • Presentation	2 Hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		6.3.4 Proof read information applying different techniques			
7. Evaluate Occupational Safety And Health Work Practices	7.1 Interpret Occupational Safety and Health practices	7.1.1 Discuss the OSH standards, principles and legislations 7.1.2 Identify OSH work practices issues 7.1.3 Discuss standard safety requirements	• Lecture • Group Discussion	• Written Exam • Demonstration • Observation • Interviews / • Questioning	1.5 hr
	7.2 Set OSH work targets	7.2.1 Discussion in actions plans that are necessary in achieving the OSH target	• Lecture • Group Discussion	• Written Exam • Demonstration • Observation • Interviews / • Questioning	1 hr
	7.3 Evaluate effectiveness of Occupational Safety and Health work instructions	7.3.1 Practice evaluating safety data (Historical or Simulated)	• Lecture • Group Discussion	• Written Exam • Demonstration • Observation • Interviews / • Questioning	1.5 hr
8.Evaluate Environmental Work Practices	8.1 Interpret Environmental practices, policies and procedures	8.1.1 Discussion Environmental Issues regarding • Water Quality • National and Local Government Issues • Safety • Endangered Species • Noise • Air Quality • Historic • Waste • Cultural	• Lecture • Group Discussion • Demonstration	• Written Exam • Demonstration • Observation • Interviews / • Questioning	1 hr

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		8.1.2 Updating of existing occupation practices			
	8.2 Establish targets to evaluate environmental practices	8.2.1 Discussion on • lower production costs and energy consumption • Environmentally Sound Processes • Resource Efficient • Recycling and Waste Management 8.2.2 Simple case study regarding energy efficiency	• Lecture • Group Discussion • Demonstration	• Written Exam • Demonstration • Observation • Interviews / Questioning	1 hr
	8.3 Evaluate effectiveness of environmental practices	8.3.1 Identifying effective environmental practices relevant to the industry/occupation • Implementation of energy efficiency	• Lecture • Group Discussion • Demonstration • Case Study	• Written Exam • Demonstration • Observation • Interviews / Questioning • Third Party Reports	1 hr
9. Facilitate Entrepreneurial Skills For Micro-Small-Medium Enterprises (MSMEs)	9.1 Develop and maintain micro-small-medium enterprise (MSMEs) skills in the organization	9.1.1 Discussions on business models and strategies 9.1.2 Discussion on Types and categories of businesses and business internal control 9.1.3 Discussion on Relevant National and local legislations affecting businesses 9.1.4 Prepare promotional materials 9.1.5 Practice basic bookkeeping	• Lecture/ Discussion • Case Study • Demonstration	• Written Test • Portfolio • Work Related Project	2 hours
	9.2 Establish and Maintain client-base/market	9.2.1 Role play on customer and employee relations	• Role Play • Lecture • Discussion	• Case problem • Written Test	2 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		9.2.2 Discussion on Basic product promotion strategies 9.2.3 Preparation of Basic Feasibility study 9.2.4 Case studies on Basic Business ethics 9.2.5 Prepare basic advertising materials	• Case study		
	9.3 Apply budgeting and financial management skills	9.3.1 Discussion on: • Basic cost-benefit analysis • Basic financial management • Basic financial accounting • Business internal controls	• Role Play • Lecture Discussion • Group work	• Written Test • Case problem	1 hour

COMMON COMPETENCIES
162 Hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
1. Validate vehicle specification	1.1 Check body type of the vehicle	• Enumerate the different kinds of vehicle • Explain the difference of each kind of vehicle • Identify the measuring points of the vehicle • Explain the procedures in measuring vehicle dimension and weight • Describe the different body shapes of the vehicle • Differentiate kinds of power train • Explain the function of each power train • Discuss occupational safety and health standard in checking the body type of a vehicle • Identify different kinds of vehicle • Measure vehicle dimensions and weight • Identify vehicle body shapes • Identify vehicle power train • Apply safety practices	• Lecture • Demonstration • Video presentation	• Written exam • Demonstrate	7 hrs
	1.2 Check vehicle engine type	• Discuss the different kinds of engine • Enumerate the different kinds of fuel/energy system • Describe the different engine components • Identify different kinds of engine	• Lecture • Demonstration • Video presentation	• Written exam • Demonstrate	3 hrs

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		- Identify different types of fuel/energy system - Identify different engine components			
	1.3 Check vehicle specifications	- Inspect VIN plate of the vehicle - Verify vehicle specification - Check vehicle modifications and conversions - Inspect vehicle conversions - Explain different vehicle related regulations in the Philippine	- Lecture - Demonstration - Video presentation	- Written exam - Demonstrate	4 hrs
	1.4 Complete validation of vehicle specification	- Explain verification of vehicle ownership using repair order and vehicle reference materials - Discuss procedures in accomplishing check sheet - Discuss submission of check sheet	- Lecture - Demonstration - Video presentation	- Written exam - Demonstrate	3 hrs
2. Move and position vehicle	2.1 Prepare vehicle for operation	- Explain vehicle multi point inspection - Enumerate cockpit drill procedure - Initialize engine startup - Perform parking brake - Show vehicle operational procedures	- Lecture discussion - Demonstration - Video presentation - Workshop visit	- Demonstration - Written exam - Interview	16 hours
	2.2 Position vehicle	- Determine workshop hazards - Discuss the procedure in avoiding workshop hazards - Define occupational health and safety standards - Move the vehicle - Explain workshop rules and regulations	- Lecture - Demonstration - Video presentation	- Demonstration - Written exam - Interview	16 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
	2.3 Park and stop the vehicle	• Explain parking rules and regulations • Park vehicle • Outline parking principles • Shut-off vehicle	• Lecture • Demonstration • Video presentation	• Demonstration • Written exam • Interview	8 hours
3. Utilize automotive tools	3.1 Prepare automotive tools	• Identify and select automotive tools and attachments • Discuss inspection and selection procedures • Describe the defects and damages of automotive tools and attachments • Discuss OSHS in preparation of automotive tools • Prepare automotive tools and attachments	• Lecture • Demonstration • Visual aids • Videos	• Written examination • Interview • Demonstration • Practical examination	6 hrs
	3.2 Use automotive tools	• Discuss the procedure in mounting attachments to automotive tools • Discuss the procedure in connecting the power supply to power tools • Discuss the procedure in operating the power tools • Discuss the utilization of hand tools • Identify PPEs • Discuss OSHS in using automotive tools • Use automotive tools • Use PPEs	• Lecture • Demonstration • Visual aids • Videos	• Written examination • Interview • Demonstration • Practical examination	6 hrs

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
	3.3 Maintain automotive tools	• Discuss the procedure in cleaning, checking for serviceability, and storing of automotive tools and attachments • Discuss the procedure in identifying and reporting defects and damages • Discuss the proper waste segregation • Demonstrate the proper maintenance of automotive tools • Demonstrate disposal of wastes	• Lecture • Visual aids • Videos	• Written examination • Demonstration	4 hrs
4. Perform mensuration and calculation	4.1 Select measuring instruments	• Describe measuring instruments • Select measuring instruments • Inspect and calibrate measuring instruments • Report and return defective measuring instruments • Demonstrate safety practices	• Demonstration • Video presentation • Lecture • Discussion • Workshop visit	• Demonstration • Written exam • Oral questioning	9 hours
	4.2 Carry out measurements and calculation	• Explain formulas for volume, areas, perimeters of plane and geometric figures • Explain the procedure in reading tools' limit of accuracy • Measure required automotive parts • Read tools' limit of accuracy • Inspect and calibrate measuring instruments	• Demonstration • Video presentation • Lecture • Discussion • Workshop visit	• Demonstration • Written exam • Oral questioning	29 hours
	4.3 Maintain measuring instruments	• Identify PPEs • Discuss cleaning procedures of measuring instruments	• Demonstration • Video presentation	• Demonstration • Written exam • Oral questioning	5 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		- Enumerate steps in storing instruments - Wear PPEs - Clean measuring instrument tools - Re-inspect and re-calibrate measuring instruments	- Lecture - Discussion		
5. Utilize workshop facilities and equipment	5.1 Perform pre-operation activities	- Identify different areas of an automotive service facilities - Explain the preparation procedures of automotive service facilities - Enumerate different equipment in the automotive service facilities - Discuss the preparation procedures of equipment - Describe minor repairs in automotive facilities and equipment - Describe defective equipment - Identify reporting procedures for defective equipment - Discuss OSHS practices related to the preparation of facilities and equipment - Prepare workshop facilities and equipment	- Lecture - Demonstration - Video presentation - Workshop visit	- Demonstration - Written exam - Interview	9 hrs
	5.2 Use facilities and equipment	- Explain the operation of equipment according to operation manual - Describe how facilities are utilized according to workshop procedures	- Lecture - Demonstration - Video presentation - Workshop visit	- Demonstration - Written exam - Interview	5 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		- Explain how equipment performance is monitored following users' manual - Describe the monitoring of facilities functionalities following workplace procedures - Discuss how OSHS safety practices are applied			
	5.3 Conduct post-operation activities	- Explain how workshop facilities are restored according to good housekeeping - Discuss tools and equipment are cleaned and stored according to good housekeeping - Explain wastes disposed following waste management procedure and OSHS - Enumerate the safety practices that are applied following OSHS - Demonstrate preparation of report based on workshop standard procedure	- Lecture - Demonstration - Video presentation - Workshop visit	- Demonstration - Written exam - Interview	5 hours
6. Prepare servicing parts and consumables	6.1 Identify parts and consumables	- Familiarize parts & consumables - Identify indirect materials - Identify hazardous parts and consumables	- Lecture - Video presentation - Actual training	- Demonstration - Written exam - Interview	6 hrs
	6.2 Retrieve and withdraw parts and consumables	- Familiarize requisition slip - Perform parts withdrawal procedure & recording - Validate parts and consumables according to quantity & specification - Perform safety precautions	- Lecture - Video presentation - Actual training	- Demonstration - Written exam - Interview	4 hrs

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
	6.3 Complete work process	• Segregate parts to be returned to customers • Segregate parts & consumables for proper disposal or recycling according to 3Rs and RA 6969 • Wear PPE's	• Lecture • Video presentation • Actual training	• Demonstration • Written exam • Interview	3 hrs
7. Prepare vehicle for servicing and releasing	7.1 Receive vehicle	• Identify different areas of an automotive service facility • Explain the receiving procedures of automotive service facilities • Explain the checklisting procedures of automotive service facilities • Describe minor repairs in automotive facilities and equipment • Discuss OSHS practices related to the preparation of facilities and equipment • Prepare workshop facilities and equipment	• Lecture • Demonstration • Video presentation • Workshop visit	• Role-playing • Written exam • Interview	6 hours
	7.2 Prepare vehicle for servicing	• Prepare vehicle for servicing • Explain the preparation procedures of automotive service facilities • Demonstrate the procedure in installing protective covers • Explain the concept of the locator blocks • Classify the type of vehicle repair based on the Repair Order	• Lecture • Demonstration	• Role-playing • Written Exams • Oral Exams	5 hours
	7.3 Prepare vehicle for releasing	• Use the repair order to identify work performed	• Lecture • Demonstration	• Role-Playing • Written Exams	3 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
		• Apply quality control measures on work done • Operate vehicle for transfer and release		• Oral Exams	

CORE COMPETENCIES
170 Hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
1. Diagnose and repair electronically controlled transmission	1.1 Prepare to diagnose and repair electronically controlled transmission	1.1.1 Determine job requirements 1.1.2 Source and interpret repair information 1.1.3 Identify work hazards 1.1.4 Manage risks 1.1.5 Select and check tools and equipment 1.1.6 Prepare report of non-serviceable tools and equipment 1.1.7 Check and report the availability of materials 1.1.8 Apply safety practices	• Lecture • Demonstration • Video presentation • Workshop visit	• Demonstration • Written exam • Interview	TOTAL = 29 Hours 3 hours
	1.2 Diagnose electronically controlled transmission	1.2.1 Carry out inspection 1.2.2 Compare inspection results 1.2.3 Report inspection findings and recommendation 1.2.4 Apply safety practices	• Lecture • Demonstration • Video presentation • Workshop visit	• Demonstration • Written exam • Interview	12 hours
	1.3 Repair electronically controlled transmission components	1.3.1 Carry out repairs 1.3.2 Carry out post-service testing 1.3.3 Apply safety practices	• Lecture • Demonstration • Video presentation • Workshop visit	• Demonstration • Written exam • Interview	12 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
	1.4 Complete work processes	1.4.1 Make final inspection base on workplace procedure 1.4.2 Turn over vehicle to immediate superior 1.4.3 Restore work area 1.4.4 Manage waste 1.4.5 Check and store tools and equipment 1.4.6 Accomplish workplace documents	• Lecture • Demonstration • Video presentation • Workshop visit	• Demonstration • Written exam • Interview	2 hours
2. Diagnose and repair continuously variable transmission	2.1 Prepare to diagnose and repair continuously variable transmission	2.1.1 Determine job requirement 2.1.2 Source and interpret diagnostic information 2.1.3 Analyze diagnostic symptoms 2.1.4 Identify hazards associated with the work 2.1.5 Select and check tools and equipment 2.1.6 Report non-serviceable tools and equipment 2.1.7 Check and report availability of materials	• Lecture • Demonstration • Video presentation • Workshop visit	• Demonstration • Written exam • Interview	TOTAL = 28 Hours 2 hours
	2.2 Diagnose continuously variable transmission	2.2.1 Perform diagnostic tests 2.2.2 Carry out inspection 2.2.3 Compare inspection results 2.2.4 Identify faults from diagnostic test 2.2.5 Report findings and recommendations 2.2.6 Apply safety practices	• Lecture • Demonstration • Video presentation • Workshop visit	• Demonstration • Written exam • Interview	12 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
	2.3 Repair continuously variable transmission components	2.3.1 Source and interpret repair information 2.3.2 Analyze repair options 2.3.3 Select and check repair tools, equipment and materials 2.3.4 Carry out repairs, replacement and components adjustments 2.3.5 Carry out post-repair testing 2.3.6 Apply safety practices	• Lecture • Demonstration • Video presentation • Workshop visit	• Demonstration • Written exam • Interview	12 hours
	2.4 Complete work processes	2.4.1 Make final inspection 2.4.2 Turned-over vehicle to immediate superior for quality control 2.4.3 Restore work area 2.4.4 Manage waste 2.4.5 Check and store tools and equipment 2.4.6 Accomplish workplace documents	• Lecture • Demonstration • Video presentation • Workshop visit	• Demonstration • Written exam • Interview	2 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
3. Diagnose and repair vehicle dynamics integrated management	3.1 Prepare to diagnose and repair vehicle dynamics integrated management	3.1.1 Determine job requirements 3.1.2 Source and interpret diagnostic and dismantling information 3.1.3 Analyze diagnostic symptoms 3.1.4 Analyze and select dismantling options 3.1.5 Identify hazards and manage risk 3.1.6 Select and check tools and equipment 3.1.7 Report non-serviceable tools and equipment 3.1.8 Check and report availability of materials	● Lecture ● Demonstration ● Video presentation ● Workshop visit	● Demonstration ● Written exam ● Interview	TOTAL = 29 Hours 3 hours
	3.2 Diagnose vehicle dynamics integrated management	3.2.1 Perform diagnostic tests 3.2.2 Carry out inspection 3.2.3 Compare inspection results 3.2.4 Identify faults and determine causes of faults 3.2.5 Report findings and recommendations 3.2.6 Apply safety practices	● Lecture ● Demonstration ● Video presentation ● Workshop visit	● Demonstration ● Written exam ● Interview	12 hours
	3.3 Repair vehicle dynamics integrated management	3.3.1 Source and interpret repair information 3.3.2 Analyze repair options 3.3.3 Select and check repair tools, equipment and materials 3.3.4 Carry out repairs, replacement and components adjustments 3.3.5 Carry out post-repair testing 3.3.6 Apply safety practices	● Lecture ● Demonstration ● Video presentation ● Workshop visit	● Demonstration ● Written exam ● Interview	12 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
	3.4 Complete work processes	3.4.1 Make final inspection 3.4.2 Turn-over vehicle to immediate superior 3.4.3 Restore work area 3.4.4 Manage waste 3.4.5 Check and store tools and equipment 3.4.6 Accomplish workplace documents	• Lecture • Demonstration • Video presentation • Workshop visit	• Demonstration • Written exam • Interview	2 hours
4. Diagnose and repair electronic 4-wheel drive control system	4.1 Identify and confirm the work requirement for electronic 4-wheel drive control system	4.1.1 Determine job requirements 4.1.2 Source and interpret diagnostic information 4.1.3 Analyze diagnostic symptoms	• Lecture • Demonstration • Video presentation • Workshop visit	• Demonstration • Written exam • Interview	TOTAL = 28 Hours 3 hours
	4.2 Prepare for diagnose and repair of electronic 4-wheel drive control system	4.2.1 Identify hazards associated with the work and manage risk 4.2.2 Select and check tools and equipment 4.2.3 Report non-serviceable tools and equipment 4.2.4 Check and report availability of materials	• Lecture • Demonstration • Video presentation • Workshop visit	• Demonstration • Written exam • Interview	3 hours
	4.3 Carry out diagnosis and determine repairs and performance enhancement strategies	4.3.1 Perform diagnostic tests 4.3.2 Carry out inspection 4.3.3 Compare inspection results 4.3.4 Identify faults and determine causes 4.3.5 Report findings and recommendations 4.3.6 Apply safety practices	• Lecture • Demonstration • Video presentation • Workshop visit	• Demonstration • Written exam • Interview	10 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
	4.4 Conduct repairs and implement performance improvement strategies	4.4.1 Interpret repair information 4.4.2 Analyze repair options 4.4.3 Select and check repair tools, equipment and materials 4.4.4 Carry out repairs and component replacements and adjustments 4.4.5 Carry post-assembly testing 4.4.6 Apply safety practices	• Lecture • Demonstration • Video presentation • Workshop visit	• Demonstration • Written exam • Interview	10 hours
	4.5 Complete work processes	4.5.1 Make final inspection 4.5.2 Turn-over vehicle to immediate superior 4.5.3 Restore work area 4.5.4 Manage waste 4.5.5 Check and store tools and equipment 4.5.6 Accomplish workplace documents	• Lecture • Demonstration • Video presentation • Workshop visit	• Demonstration • Written exam • Interview	2 hours
5.Diagnose and repair 4-wheel drive assembly	5.1 Prepare to diagnose and repair 4-wheel drive assembly	5.1.1 Determine job requirements 5.1.2 Source and interpret diagnostic and dismantling information 5.1.3 Analyze diagnostic symptoms 5.1.4 Analyze and select dismantling options 5.1.5 Identify hazards and manage risk 5.1.6 Select and check tools and equipment 5.1.7 Report non-serviceable tools and equipment 5.1.8 Check and report availability of materials	• Lecture • Demonstration • Video presentation • Workshop visit	• Demonstration • Written exam • Interview	TOTAL = 28 hrs 2 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
	5.2 Diagnose 4-wheel drive assembly	5.2.1 Perform diagnostic tests 5.2.2 Carry out inspection 5.2.3 Compare inspection results 5.2.4 Identify faults and determine causes 5.2.5 Report findings and recommendations 5.2.6 Apply safety practices	• Lecture • Demonstration • Video presentation • Workshop visit	• Demonstration • Written exam • Interview	12 hours
	5.3 Repair 4-wheel drive assembly	5.3.1 Source and interpret repair information 5.3.2 Analyze repair options 5.3.3 Select and check repair tools, equipment and materials 5.3.4 Carry out repairs, replacement and components adjustments 5.3.5 Carry out post-repair testing 5.3.6 Apply safety practices	• Lecture • Demonstration • Video presentation • Workshop visit	• Demonstration • Written exam • Interview	12 hours
	5.4 Perform activities to completion	5.4.1 Make final inspection 5.4.2 Turn-over vehicle to immediate supervisor 5.4.3 Restore work area 5.4.4 Manage waste 5.4.5 Check and store tools and equipment 5.4.6 Accomplish workplace documents	• Lecture • Demonstration • Video presentation • Workshop visit	• Demonstration • Written exam • Interview	2 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
6. Diagnose complex faults in electric power steering	6.1 Identify and confirm the work requirement of electric power steering	6.1.1 Determine job requirements 6.1.2 Source and interpret diagnostic information 6.1.3 Analyze diagnostic symptoms 6.1.4 Identify hazards associated with the work 6.1.5 Select and check tools, equipment and materials	• Lecture • Demonstration • Video presentation • Workshop visit	• Demonstration • Written exam • Interview	TOTAL = 28 Hours 2 hours
	6.2 Prepare to perform diagnosis for electric power steering	6.2.1 Perform diagnostic tests 6.2.2 Carry out Inspection 6.2.3 Compare inspection results 6.2.4 Identify faults and determine causes of faults 6.2.5 Report findings 6.2.6 Apply safety practices	• Lecture • Demonstration • Video presentation • Workshop visit	• Demonstration • Written exam • Interview	12 hours
	6.3 Apply diagnostic procedures for electric power steering	6.3.1 Source and interpret repair information 6.3.2 Analyze repair options 6.3.3 Select and check repair tools, equipment and materials 6.3.4 Carry out repairs and component replacements and adjustments 6.3.5 Carry out post-service testing 6.3.6 Apply safety practices	• Lecture • Demonstration • Video presentation • Workshop visit	• Demonstration • Written exam • Interview	12 hours

Unit of Competency	Learning Outcomes	Learning Activities	Methodology	Assessment Approach	Nominal Duration
	6.4 Complete work processes	6.4.1 Make final inspection 6.4.2 Turn-over vehicle to immediate superior 6.4.3 Restore work area 6.4.4 Manage waste 6.4.5 Check and store tools and equipment 6.4.6 Accomplish workplace documents	• Lecture • Demonstration • Video presentation • Workshop visit	• Demonstration • Written exam • Interview	2 hours

3.2 TRAINING DELIVERY

- 1. The delivery of training shall adhere to the design of the curriculum. Delivery shall be guided by the principles of competency-based TVET.**
 - a. Course design is based on competency standards set by the industry or recognized industry sector; (Learning system is driven by competencies written to industry standards)
 - b. Training delivery is learner-centered and should accommodate individualized and self-paced learning strategies;
 - c. Training can be done on an actual workplace setting, simulation of a workplace and/or through adoption of modern technology.
 - d. Assessment is based in the collection of evidence of the performance of work to the industry required standards;
 - e. Assessment of competency takes the trainee's knowledge and attitude into account but requires evidence of actual performance of the competency as the primary source of evidence.
 - f. Training program allows for recognition of prior learning (RPL) or current competencies;
 - g. Training completion is based on satisfactory performance of all specified competencies.
- 2. The competency-based TVET system recognizes various types of delivery modes, both on-and off-the-job as long as the learning is driven by the competency standards specified by the industry. The following training modalities and their variations/components may be adopted singly or in combination with other modalities when designing and delivering training programs:**

2.1 School/Institution- Based:

- Dual Training System (DTS)/Dualized Training Program (DTP) which contain both in-school and in-industry training or fieldwork components. Details can be referred to the Implementing Rules and Regulations of the DTS Law and the TESDA Guidelines on the DTP.
- Distance learning is a formal education process in which majority of the instruction occurs when the students and instructor are not in the same place. Distance learning may employ correspondence study, audio, video, computer technologies or other modern technologies that can be used to facilitate learning and formal and non-formal training. Specific guidelines on this mode shall be issued by the TESDA Secretariat.
- Supervised Industry Training (SIT) or on-the-job training (OJT) is an approach in training designed to enhance the knowledge and skills of the trainee through actual experience in the workplace to acquire specific competencies as prescribed in the training regulations. It is imperative that the deployment of trainees in the workplace is

adhered to training programs agreed by the institution and enterprise and status and progress of trainees are closely monitored by the training institutions to prevent opportunity for work exploitation.

- The classroom-based or in-center instruction uses of learner-centered methods as well as laboratory or field-work components.

2.2 Enterprise-Based:

- **Formal Apprenticeship** – Training within employment involving a contract between an apprentice and an enterprise on an approved apprenticeable occupation.
- **Informal Apprenticeship** - is based on a training (and working) agreement between an apprentice and a master craftsperson wherein the agreement may be written or oral and the master craftsperson commits to training the apprentice in all the skills relevant to his or her trade over a significant period of time, usually between one and four years, while the apprentice commits to contributing productively to the work of the business. Training is integrated into the production process and apprentices learn by working alongside the experienced craftsperson.
- **Enterprise-based Training** - where training is implemented within the company in accordance with the requirements of the specific company. Specific guidelines on this mode shall be issued by the TESDA Secretariat.

2.3 Community-Based – short term program conducted by non-government organizations (NGOs), LGUs, training centers and other TVET providers which are intended to address the specific needs of a community. Such programs can be conducted in informal settings such as barangay hall, basketball courts, etc. These programs can also be mobile training program (MTP)

3.3 TRAINEE ENTRY REQUIREMENTS

Trainees or students who would like to enroll in this program must possess the following requirements:

- Must be holder of Automotive Servicing (Chassis Repair) NC II;
- Can communicate both oral and written;
- Can perform basic mathematical computation;
- Must be able to drive a vehicle; and
- Must possess a valid driver license.

This list does not include specific institutional requirements such as educational attainment, appropriate work experience, and others that may be required of the trainees by the school or training center delivering the TVET program.

3.4 TOOLS, EQUIPMENT AND MATERIALS

AUTOMOTIVE DIAGNOSIS (CHASSIS) NC III

Recommended list of tools, equipment and materials for the training of 25 trainees for Automotive Diagnosis (Chassis) NC III.

Up-to-date tools, materials, and equipment of equivalent functions can be used as alternatives.

TOOLS	
QTY	DESCRIPTION
4 sets	Standard technician hand tools
1 unit	Diagnostic scan tools (OBD 2)
2 pcs	Dial gauge with magnetic base
4 pcs	Vernier caliper
1 pcs	Torque wrench (30 kgf- cm)
1 pcs	Torque wrench (230 kgf- cm)
1 pcs	Torque wrench (460 kgf- cm)
1 pcs	Torque wrench (920 kgf- cm)
1 pcs	Torque wrench (1900 kgf- cm)
1 pcs	Torque wrench (2800 kgf- cm)
5 pcs	Plug (prevent oil leakage from transmission)
2 pcs	Multi-meter tester (Digital)
1 pc	Oil pressure gauge set
1 pc	Engine lifting brackets (SST)
1 pc	Engine support set (SST)
1 pc	Engine support hanger (SST)
1 pc	Ring gear brake set (SST)
2 pcs	Oil seal installer (SST)
2 pcs	Dust boot installer (SST)
4 pcs	V block

TOOLS	
QTY	DESCRIPTION
4 pcs	Wheel chocks
4 pcs	Snap ring expander
1 pc	Vacuum gauge set (SST)
1 pc	Thickness gauge
4 pcs	Pin punch (5mm)*
4 pcs	Plastic hammer
5 pcs	Ball peen hammer, 1lb
4 pcs	Micrometer (0 – 25mm)
4 pcs	Caliper gauge
4 pcs	Inside diameter measuring tools
1 set	Drill bit set
1 set	Screw extractor set
2 pcs	Steering wheel puller
2 pcs	Torx driver (6mm)
2 pcs	Torx driver (8mm)
4 pcs	Rack & pinion holder
4 pcs	Steering adjustable wrench
4 pcs	Hexagon wrench (24mm)
1 pc	Ball joint puller set (SST)
1 pc	Bearing puller set (SST)
4 pcs	Stainless Steel Ruler, 12"
1 pc	Bearing installer (SST)
1 pc	Support block (SST)
1 pc	Boot clamp crimpers
1 pc	Dust boot installer (SST)
1 pc	Oil pressure gauge set with adapters
1 set	Plastic clip remover
1 pc	Grease gun
1 pc	Tire gauge
5 pcs	Trouble light
5 pcs	Feeler gauge
2 pcs	C-clamp
1 pc	Bearing and oil seal remover (SST)
4 pcs	Brake bleeder
1 pc	Tire pressure gauge, ball pen type
2 pcs	Push-pull scale
2 pcs	Vinyl hose with clear container
2 pcs	SST (puller for removing transmission gears)*
2 pcs	SST (to remove flange coupling lock nut)*
2 pcs	SST (to remove flange coupling)*
2 pcs	Special tools (such as sliding hammer)*
5 pcs	Soldering iron

EQUIPMENT	
QTY	DESCRIPTION
1 unit	Training vehicle with (complete with Electronically Controlled Transmission) ECT, anti-lock brake system, electronic stability control and traction control system, hydraulic controlled brake system, electronic 4-wheel drive control system, and electric power steering) Continuously variable transmission component
1 unit	Vehicle Lifter, 2-post, 3Tons
1 unit	Air reel
1 unit	Electrical reel
1 unit	Workshop air compressor with air line, 2hp
1 unit	Electric drill
5 units	Air dust gun
4 units	Work bench with vise
1 unit	Hydraulic press, 1 ton
1 unit	Bench Grinder, 1hp
1 unit	Tracking gauge for measuring toe angle
2 units	Turning radius gauge
1 unit	Crocodile jack, 3Tons
1 unit	Wheel balancer
1 unit	Grease gun
8 pcs	Wheel wedge
1 pc	Creeper
1 pc	Trouble light
1 pc	Oil bucket, 20L cap.
1 pc	Drain pan, 20L
1 pc	Differential stand
1 unit	Wheel aligner
4 unit	Car jack stands
1 unit	Transmission jack
1 unit	Brake lathe machine
1 unit	ABS wheel sensor
1 unit	Bench vise

MATERIALS	
QTY	DESCRIPTION
5 cans	Grease (500mg/can)
2 cans	Penetrating oil (500ml/cans)
25 pcs	Rags
8 L	Gear oil
8 L	Engine oil
2 L	Brake fluid
5 set	Mop
5 units	Trash bins, 10 L
2 rolls	Electrical tape
100 grams	lithium soap base glycol grease
4 sets	Caliper brake overhauling (O/H) kit

MATERIALS	
QTY	DESCRIPTION
8 L	Electronically controlled transmission fluid
8 L	Continuously variable transmission fluid
2 sets	Shock absorbing foam
12 pcs	Sandpaper, assorted
2 pcs	Tailor Chalk
4 pcs	Container (for bleeding fluid)
4 pcs	Tray, aluminum, 20cmx20cm
4 pcs.	Vinyl hose (2 m)
10 pcs	Adjusting bolt (of various sizes)*
1 Kilo	Grease
1 tube	Silicon sealant
1 tube	Mechanic blue, 200ml
1 box	Break cleaner
1 can	Degreaser, 450ml
4 copies	Manuals/ References: -Repair manuals -Wiring Diagrams -Workshop
	PPEs:
25 pair	Gloves
10 pcs	Hard hat
25 pairs	Safety shoes
25 pcs	Cover all
25 pcs	Dust mask
1 set	First aid kit
1 unit	Fire extinguisher
5 sachets	Soap suds
25 pcs	Goggles
1 kit	Vehicle protective - seat cover - steering wheel cover - fender cover - shifting knob / lever cover - paper matting

*Note: Please refer to repair manual of specific brand and model.

List of tools is based on Mazda CX-9 model.

**Whichever is available in the locality.

Note: Access to and use of equipment and facilities can be provided through cooperative arrangements of MOA with other partner enterprises/organizations/institutions.

3.5 TRAINING FACILITIES

AUTOMOTIVE DIAGNOSIS (CHASSIS) NC III

Based on a class intake of 25 learners/trainees.

SPACE REQUIREMENT	SIZE IN METERS	AREA IN SQ. METERS	TOTAL AREA IN SQ. METERS
A. Building (permanent)			
• Lecture Room	5 x 6	30.00	30.00
• Workshop/ Laboratory area (Work bay)	4 x 7	28 x 2	56.00
• Storage Area (Tool room & S/M storage area)		20	20.00
• Learning Resource Center	5 x 4	20.00	20.00
• Wash area/ comfort room (Male, Female, PWD)		12	12.00
TOTAL			138.00

Note: Access to and use of equipment and facilities can be provided through cooperative arrangements of MOA with other partner enterprises/organizations/institutions.

3.6 TRAINERS' QUALIFICATIONS

NEW TRAINER

- Holder of National TVET Trainers Certificate (NTTC) Level 1 in-Automotive Diagnosis (Chassis) NC III or Automotive Servicing NC IV; and
- Must have at least 1-year industry experience on Automotive Diagnosis within the last 3 years.

EXISTING TRAINER

- Holder of National TVET Trainers Certificate (NTTC) Level 1 in Automotive Diagnosis (Chassis) NC III or Automotive Servicing NC IV; and
- Must have industry immersion of 80 hours annually (industry training which includes structured training program inclusive of hands-on activities, observation in a workshop, and training certificates with number of hours).

3.6 INSTITUTIONAL ASSESSMENT

Institutional Assessment is gathering of evidence to determine the achievements of the requirements of the qualification to enable the trainer to make judgment whether the trainee is competent or not yet competent.

SECTION 4 NATIONAL ASSESSMENT AND CERTIFICATION ARRANGEMENTS

Competency Assessment is the process of collecting evidence and making judgments whether competency has been achieved. The purpose of assessment is to confirm that an individual can perform to the standards expected at the workplace as expressed in relevant competency standards.

The assessment process is based on evidence or information gathered to prove achievement of competencies. The process may be applied to a full qualification or employable unit(s) of competency in partial fulfillment of the requirements of the national qualification.

4.1 NATIONAL ASSESSMENT AND CERTIFICATION ARRANGEMENTS

- 4.1.1 To attain the national qualification of AUTOMOTIVE DIAGNOSIS (CHASSIS) NC III, the candidate must demonstrate competence in all units listed in Section 1. Successful candidates shall be awarded a National Certificate signed by the TESDA Director General.
- 4.1.2 Assessment shall cover all competencies with basic and common integrated or assessed concurrently with the core units of competency.
- 4.1.3 The following are qualified to apply for assessment and certification, as long as they are holders of National Certificate in the amended Automotive Servicing (Chassis Repair) NC II:
 - 4.1.3.1 Graduates of WTR-registered program on Automotive Diagnosis (Chassis) NC III, or graduates of NTR programs or formal/non-formal/informal including enterprise-based training programs related to automotive servicing (chassis repair); or
 - 4.1.3.2 Candidates who gained competencies in implementing automotive chassis diagnosis on vehicle through informal training or previous work experiences. For informal training, a certification from any government agency or LGU or Barangay indicating the competencies of an individual. For work experience, a certificate of Employment and Job Description must be provided as proof.
- 4.1.4 Current holders of National Certificate (NC) and Certificate of Competency (COC) in AUTOMOTIVE SERVICING NC III shall have to undergo reassessment in the amended Training Regulations upon expiration of their Certificates. Candidate must be a holder of National Certificate in the amended Automotive Servicing (Chassis Repair) NC II.
- 4.1.5 Recognition of Prior Learning (RPL). Candidates who have gained competencies through informal training, previous work or life experiences may apply for recognition in a particular qualification through competency based on the current TESDA circular and/or implementing guidelines on portfolio assessment and RPL.

- 4.1.6 The industry shall determine assessment and certification requirements for each qualification with promulgated Training Regulations. It includes the following:
- a. entry requirements for candidates
 - b. evidence gathering methods
 - c. qualification requirements of competency assessors
 - d. specific assessment and certification arrangements as by industry.

4.2 Competency Assessment Requisite

- 4.2.1 Self-Assessment Guide. The self-assessment guide (SAG) is accomplished by the candidate prior to actual competency assessment. SAG is a pre-assessment tool to help the candidate and the assessor determine what evidence is available, where gaps exist, including readiness for assessment.

This document can:

- a. Identify the candidate's skills and knowledge
 - b. Highlight gaps in candidate's skills and knowledge
 - c. Provide critical guidance to the assessor and candidate on the evidence that need to be presented
 - d. Assist the candidate to identify key areas in which practice is needed or additional information or skills that should be gained prior
- 4.2.2 Accredited Assessment Center. Only Assessment Center accredited by TESDA is authorized to conduct competency assessment. Assessment centers undergo a quality assured procedure for accreditation before they are authorized by TESDA to manage the assessment for National Certification.
- 4.2.3 Accredited Competency Assessor. Only accredited competency assessor is authorized to conduct assessment of competence. Competency assessors undergo a quality assured system of accreditation procedure before they are authorized by TESDA to assess the competencies of candidates for National Certification.

COMPETENCY MAP

AUTOMOTIVE DIAGNOSIS (CHASSIS) NC III

BASIC COMPETENCIES

Receive and respond to workplace communication	Participate in workplace communication	Lead workplace communication	Utilize specialized communication skill	Manage and sustain effective communication strategies
Work with others	Work in a team environment	Lead small teams	Develop and lead teams	Manage and sustain high performing teams
Solve/address routine problems	Solve/address general workplace problems	Apply critical thinking and problem-solving techniques in the workplace	Perform higher-order thinking processes and apply techniques in the workplace	Evaluate higher order thinking skills and adjust problem solving techniques
Enhance self-management skills	Develop career and life decisions	Work in a diverse environment	Contribute to the practice of social justice in the workplace	Advocate strategic thinking for global citizenship
Support innovation	Contribute to workplace innovation	Propose methods of applying learning and innovation in the organization	Manage innovative work instructions	Incorporate innovation into work procedures
Access and maintain information	Present relevant information	Use information systematically	Manage and evaluate usage of information	Develop systems in managing, and maintaining information
Follow occupational safety and health policies and procedures	Practice occupational safety and health policies and procedures	Evaluate occupational safety and health work practices	Lead in improvement of occupational safety and health program, policies and procedures	Manage implementation of OSH programs in the workplace
Apply environmental work standards	Exercise efficient and effective sustainable practices in the workplace	Evaluate environmental work practices	Lead towards improvement of environmental work programs, policies and procedures	Manage implementation of environmental programs in the workplace
Adopt entrepreneurial mindset in the workplace	Practice entrepreneurial skills in the workplace	Facilitate entrepreneurial skills for micro-small-medium enterprises (MSMEs)	Sustain entrepreneurial skills	Develop and sustain a high-performing enterprise

COMMON COMPETENCIES

Validate vehicle specification	Move and position vehicle	Utilize automotive tools	Perform mensuration and calculation	Utilize workshop facilities and equipment
Prepare servicing parts and consumables	Prepare vehicle for servicing and releasing			

CORE COMPETENCIES

Diagnose and repair of electronically controlled transmission	Diagnose and repair of continuously variable transmission	Diagnose and repair of vehicle dynamics integrated management	Diagnose and repair of electronic 4-wheel drive control system	Diagnose and repair 4-wheel drive assembly
Diagnose and repair electric power steering				

GLOSSARY OF TERMS

1. Adjustment	a small alteration or movement made to achieve a desired fit, appearance, or result.
2. Diagnose	identify the nature of problem by inspection of the symptoms.
3. Diagnostic symptoms	a physical manifestation that is regarded as indicating a condition of malfunction.
4. Evaluation of components	the making of a judgment about the condition of a part/component.
5. Final inspection	Includes road testing, oil leakage and functionality
6. Initialization	The process of the user specifying initialization valued is sometimes called configuration
7. Maintenance	the regular or periodic maintenance servicing of vehicles to keep it in top condition.
8. OSHS	Occupational Safety and Health Standard
9. Out of standard	worn-out, unserviceable components, not conforming to manufacturer's standard.
10.Overhaul	take apart a major automobile component in order to examine it and repair/replace a part if necessary to bring back the major component to working condition.
11.Repair	fix or return to working condition a part/component. It refers to cleaning, adjustment, and replacement.
12. Service	the act of rendering maintenance service and repair/replacement of parts of an automobile to keep it in top condition.
13.Wheel alignment	part of standard automobile maintenance that consists of adjusting the angles of wheels so that they conform to manufacturer's standard.
14.Wheel balance	describes the distribution of mass within an automobile tire or the entire wheel (including the rim) to which it is attached.



TRAINING REGULATIONS (TR) DOCUMENT REVISION HISTORY

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Qualification Code: **ALTADC324**

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